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Extratropical transition of tropical cyclones over the Southern Hemisphere

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29th April 2026

Extratropical transition of tropical cyclone

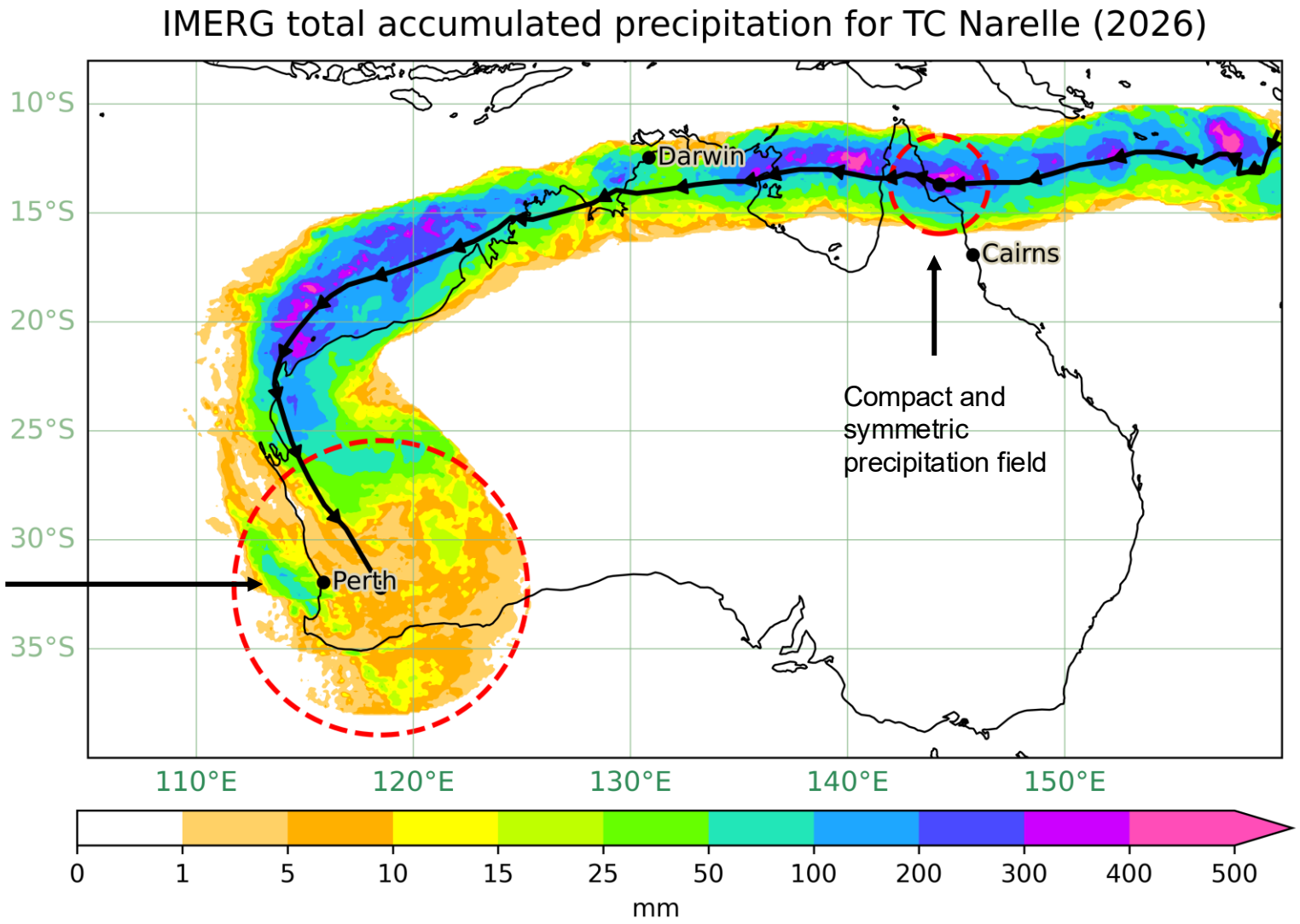
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Tropical cyclones that move into the midlatitudes undergo changes in their structure and transition into extratropical cyclones. The process is known as extratropical transition (ET).

ET can lead to severe weather locally and affect weather downstream.

Changes in precipitation field during ET

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Downstream impacts of ET

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Rossby wave packets downstream TC recurvature

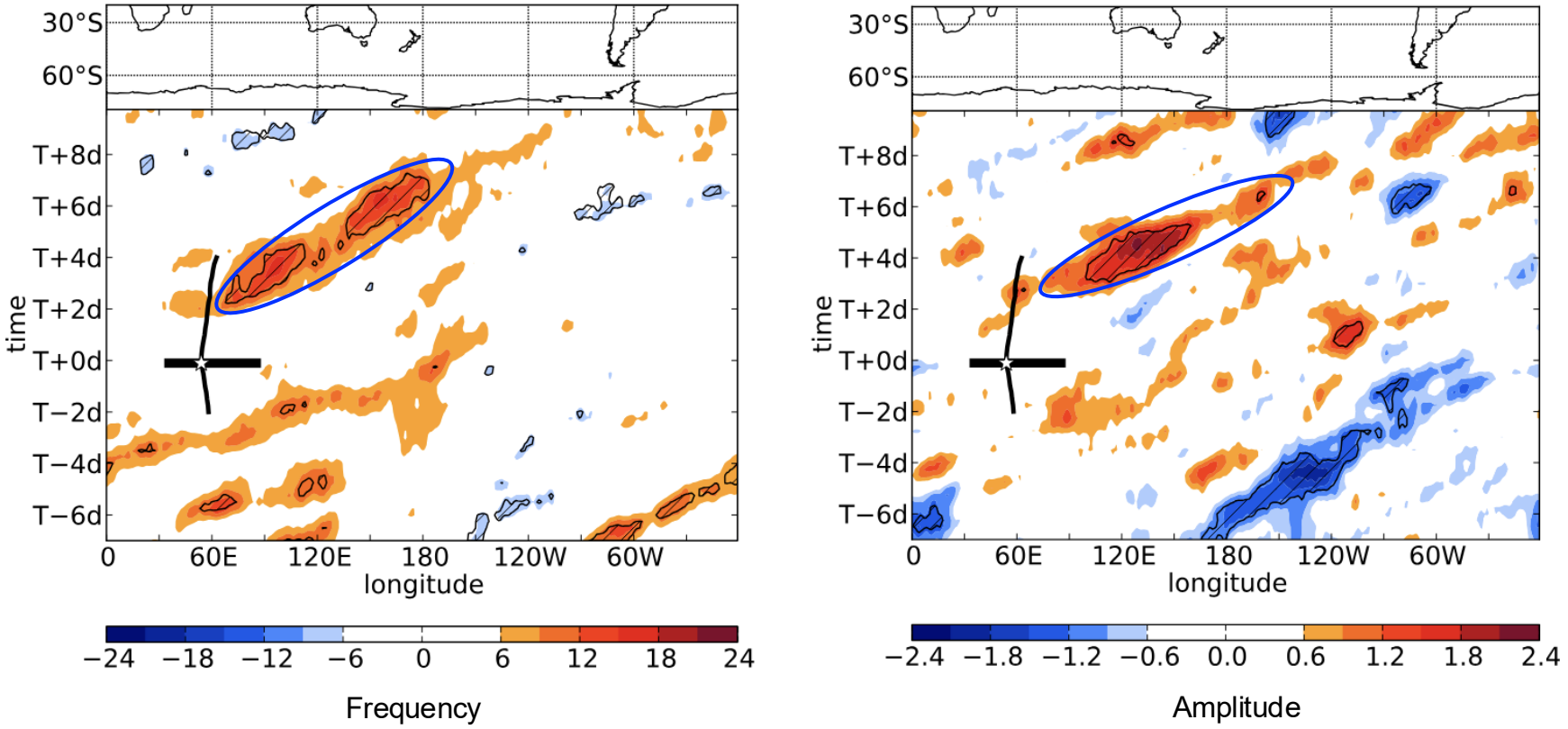
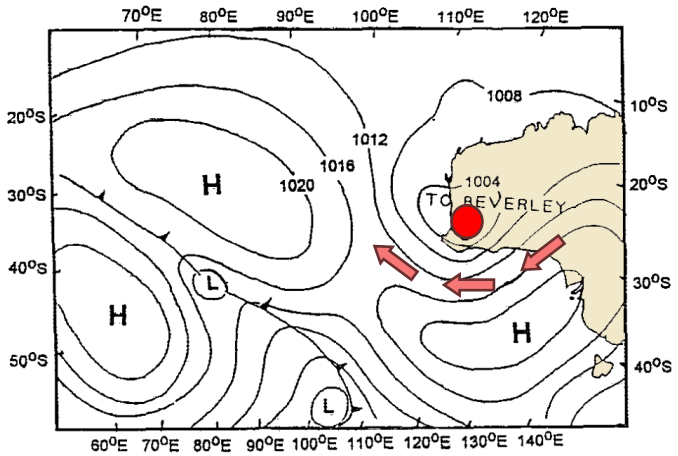


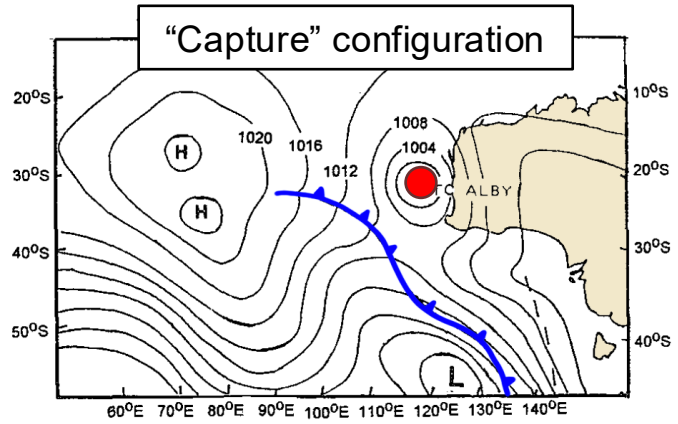
Figure adapted Quinting et al. (2016)

Limited ET studies in the Southern Hemisphere

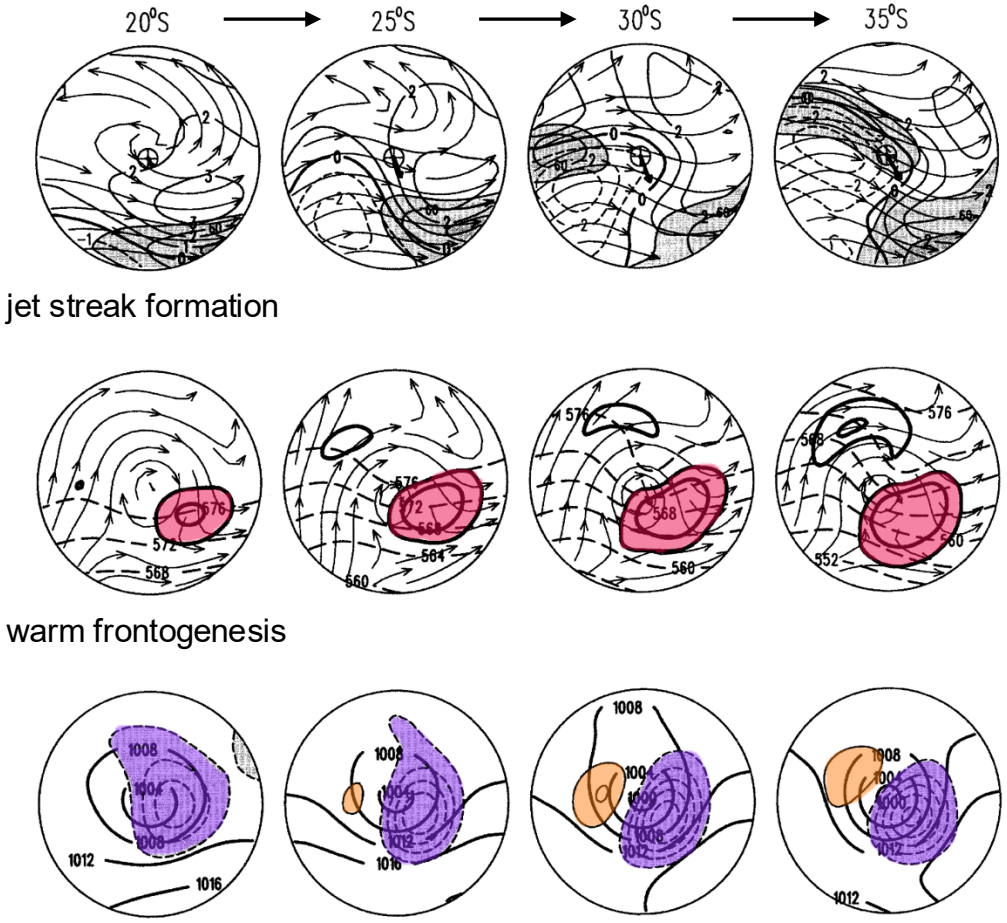
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- Foley and Hanstrum (1994) – northwest Australia





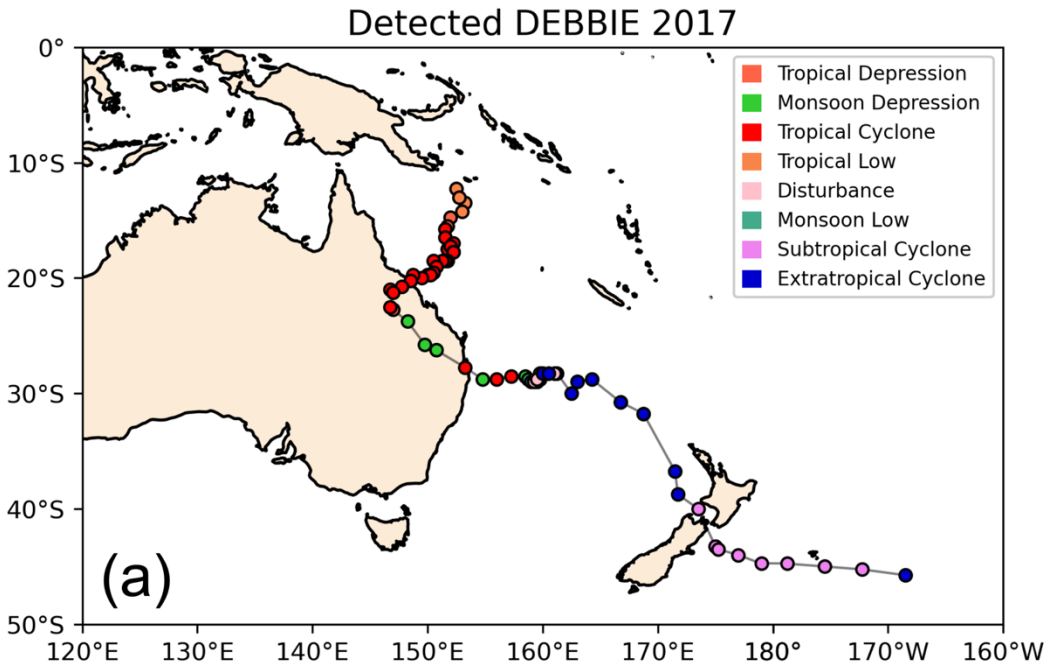
- Sinclair (2002, 2004) – southwest Pacific

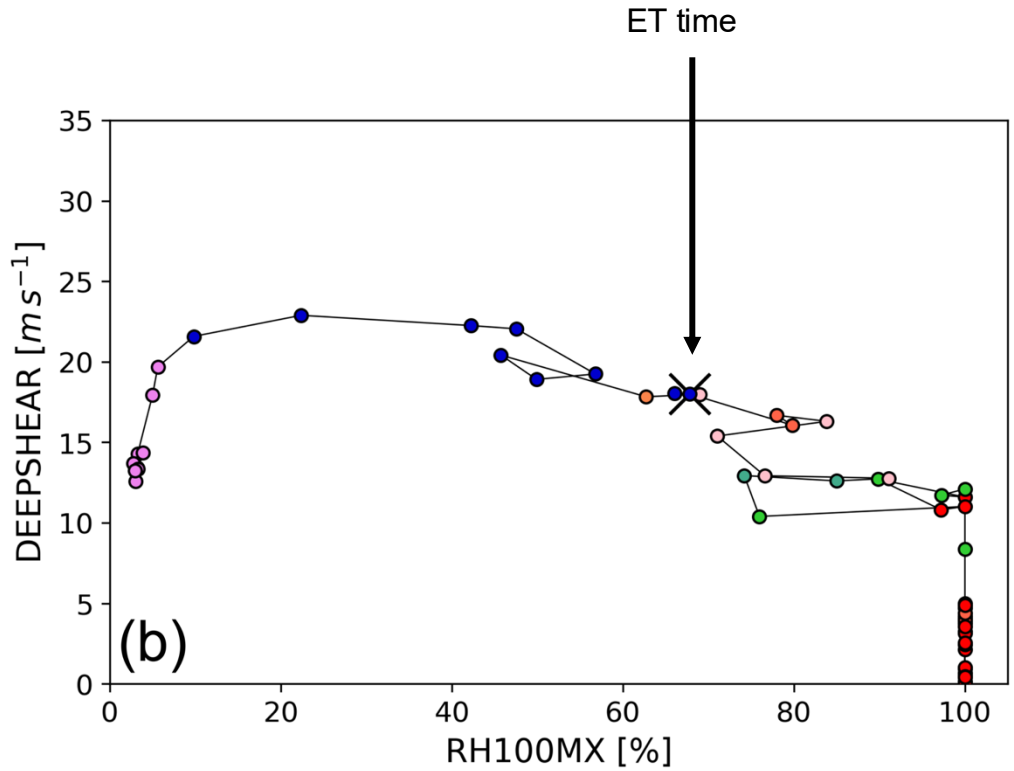


Figures adapted from Foley and Hanstrum (1994) and Sinclair (2002)

Objective ET identification

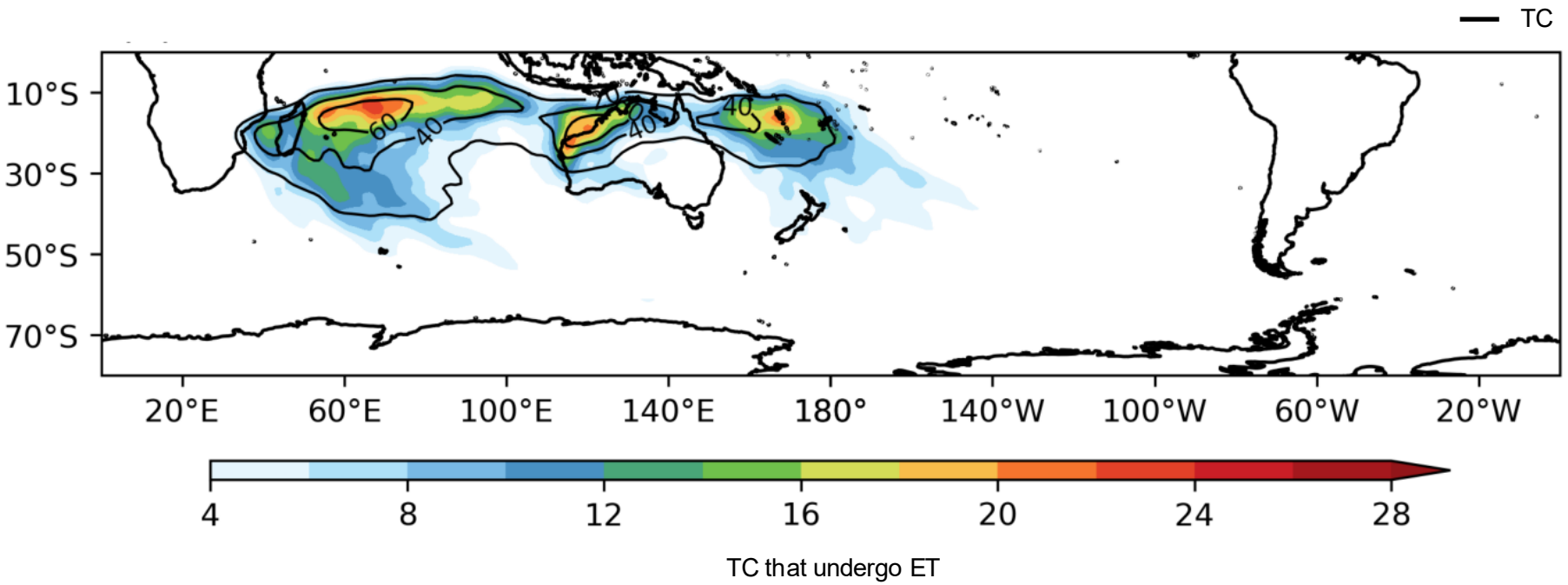
System for Classification of Low-Pressure Systems (SyCLoPS; Han and Ullrich, 2025) on ERA5 reanalysis





The RH100MX-DEEPSHEAR method identifies 506 ET events out of a total 984 TC tracks over the Southern Hemisphere during 1979-2021 period

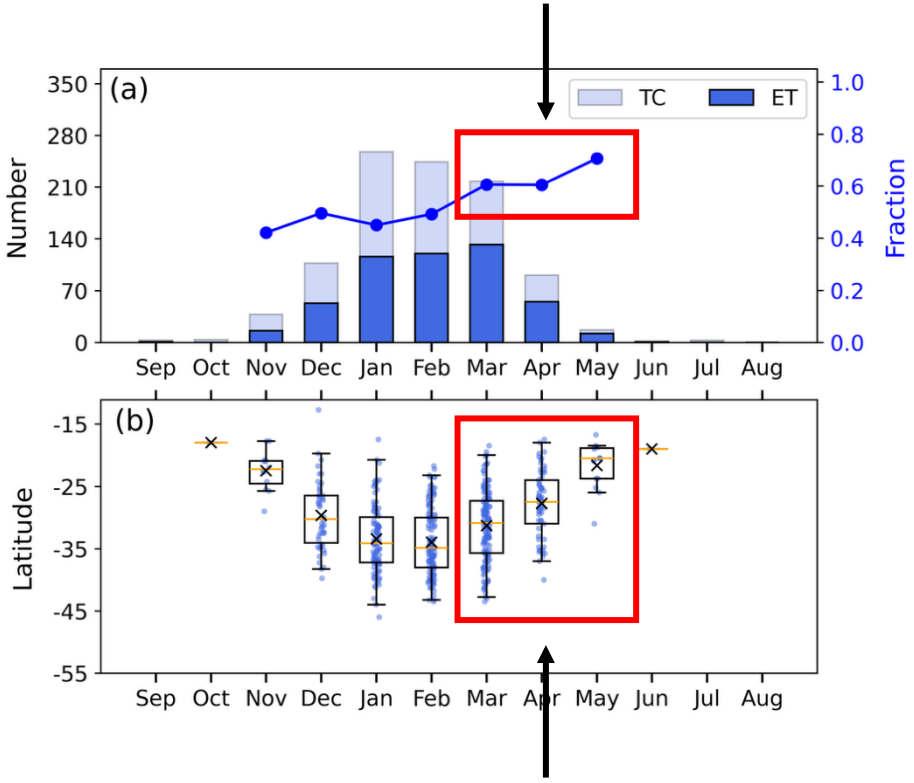
Climatology of detected ET events



Seasonality of detected ET events

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Higher ET frequency in austral autumn

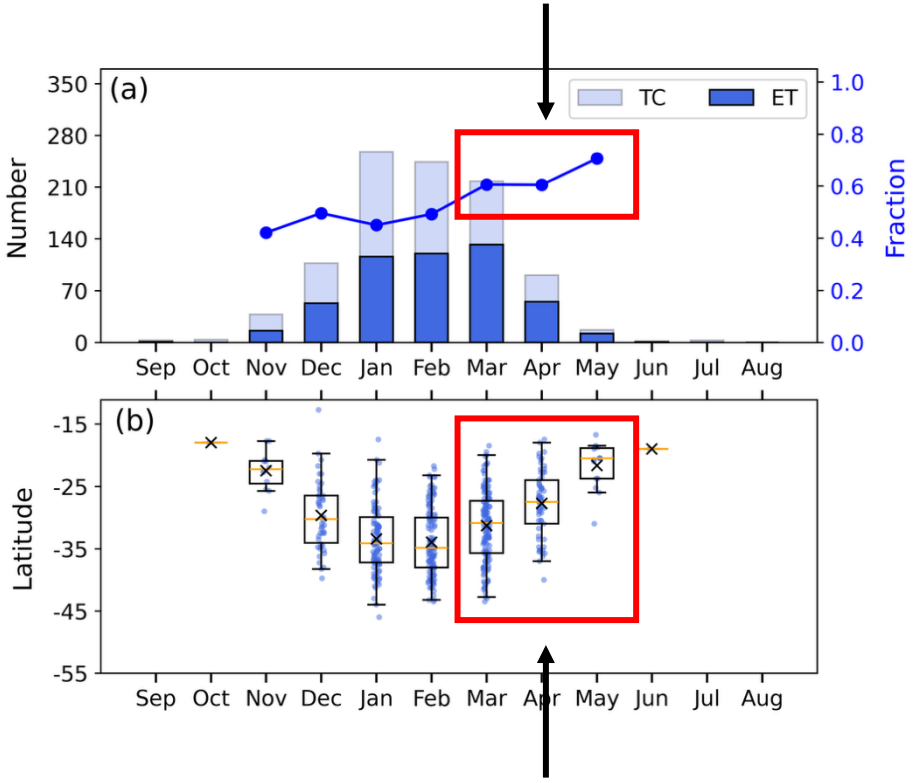


ET events occur at lower latitudes

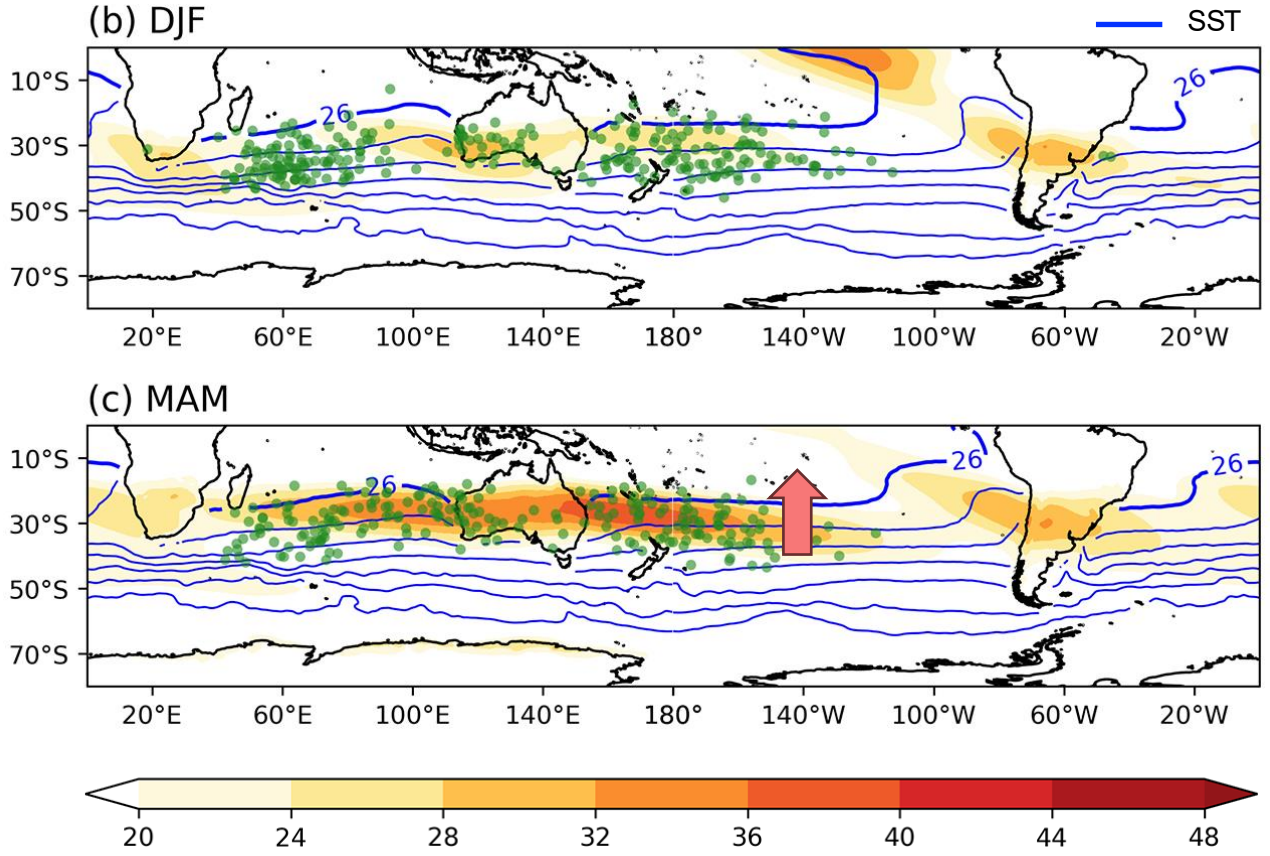
Seasonality of detected ET events

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Higher ET frequency in austral autumn

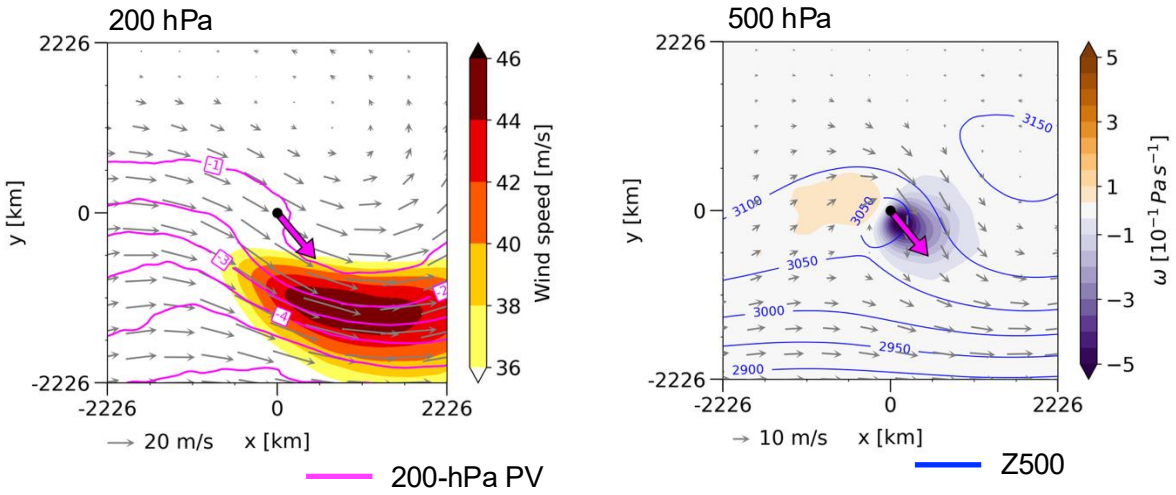


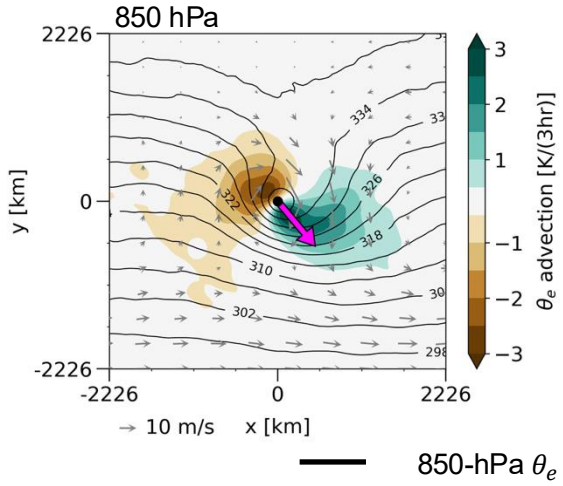
ET events occur at lower latitudes



200-850 hPa vertical wind shear (m/s)

Mean characteristics of ET

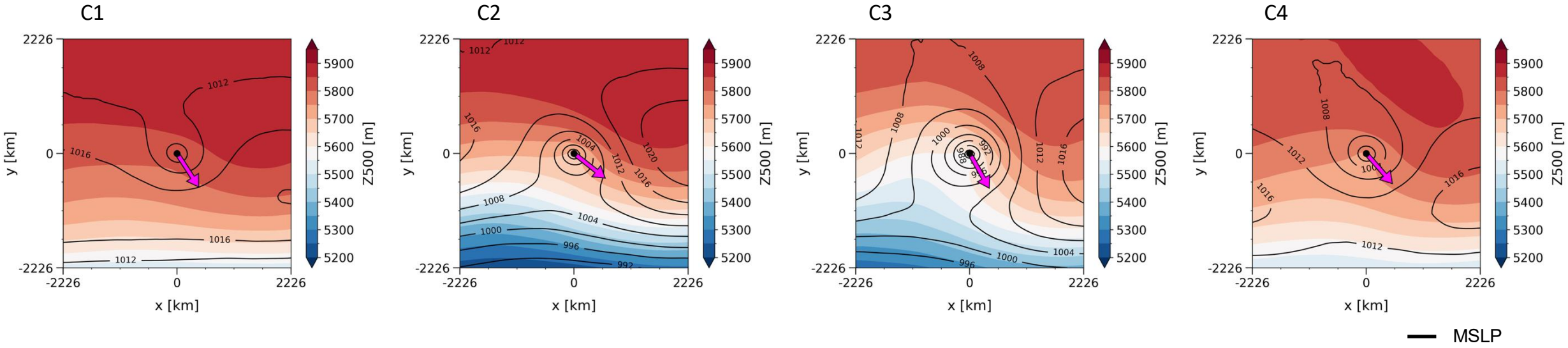


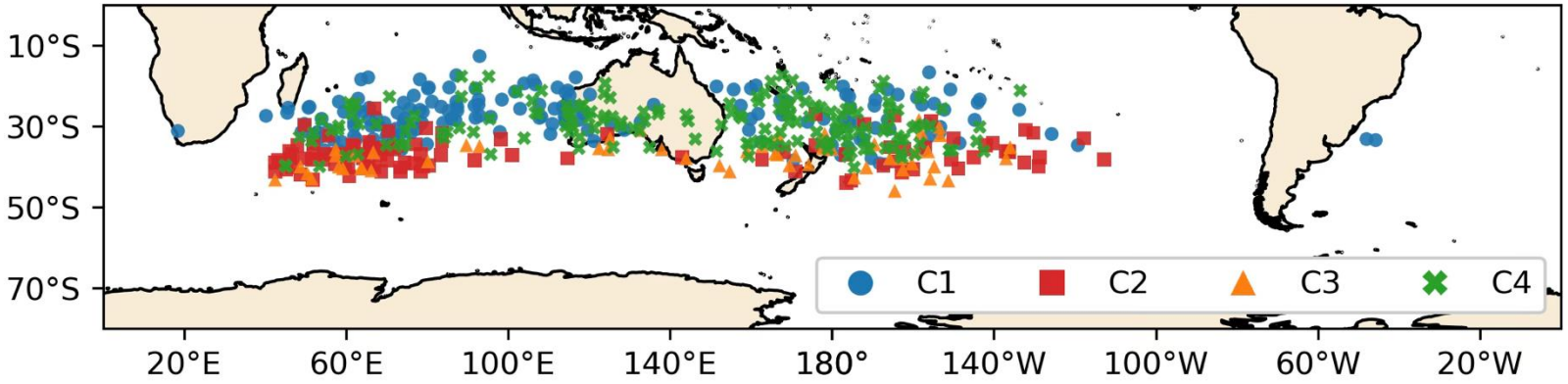


- 200 hPa: jet streak formation
- 500 hPa: asymmetric vertical motion
- 850 hPa: enhanced thermal gradient

Clustering on synoptic configuration of ET

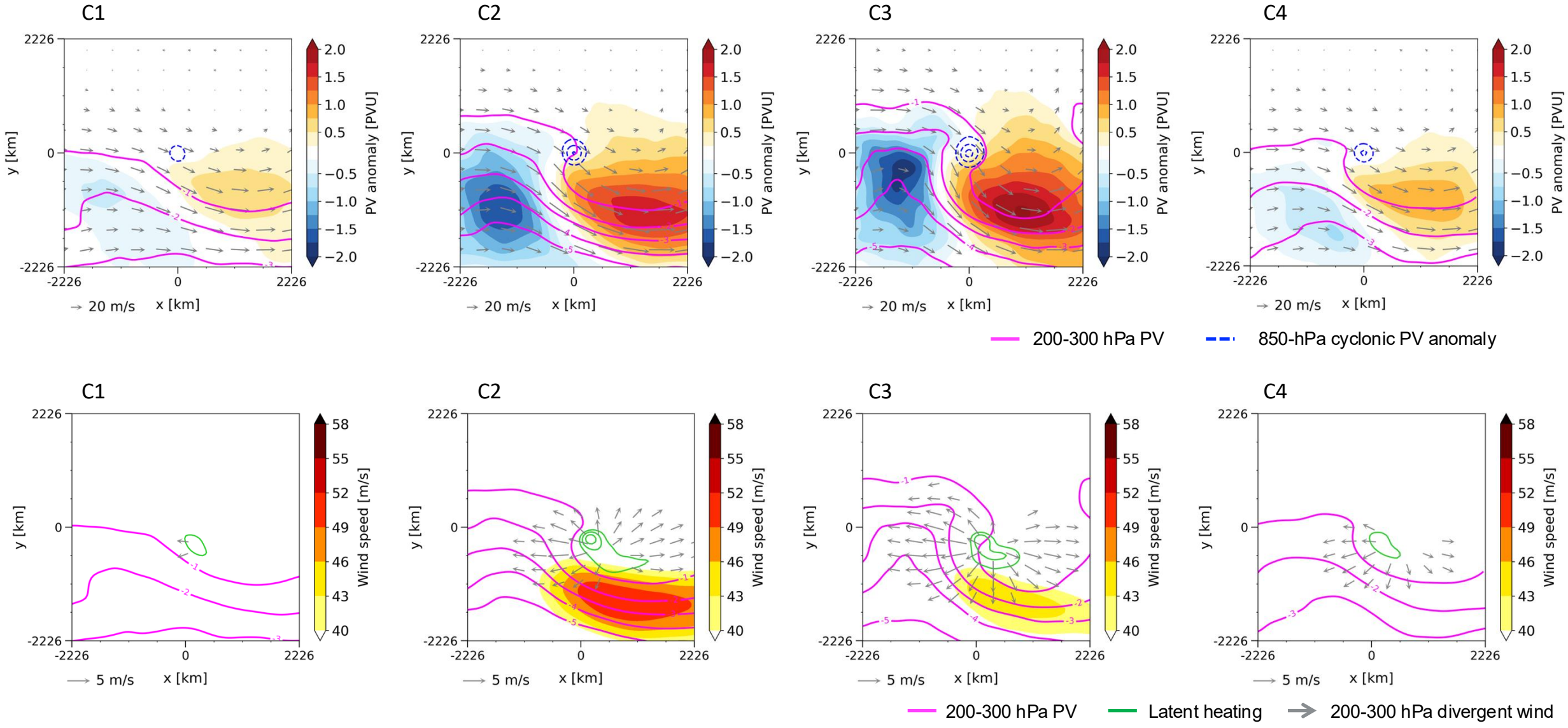
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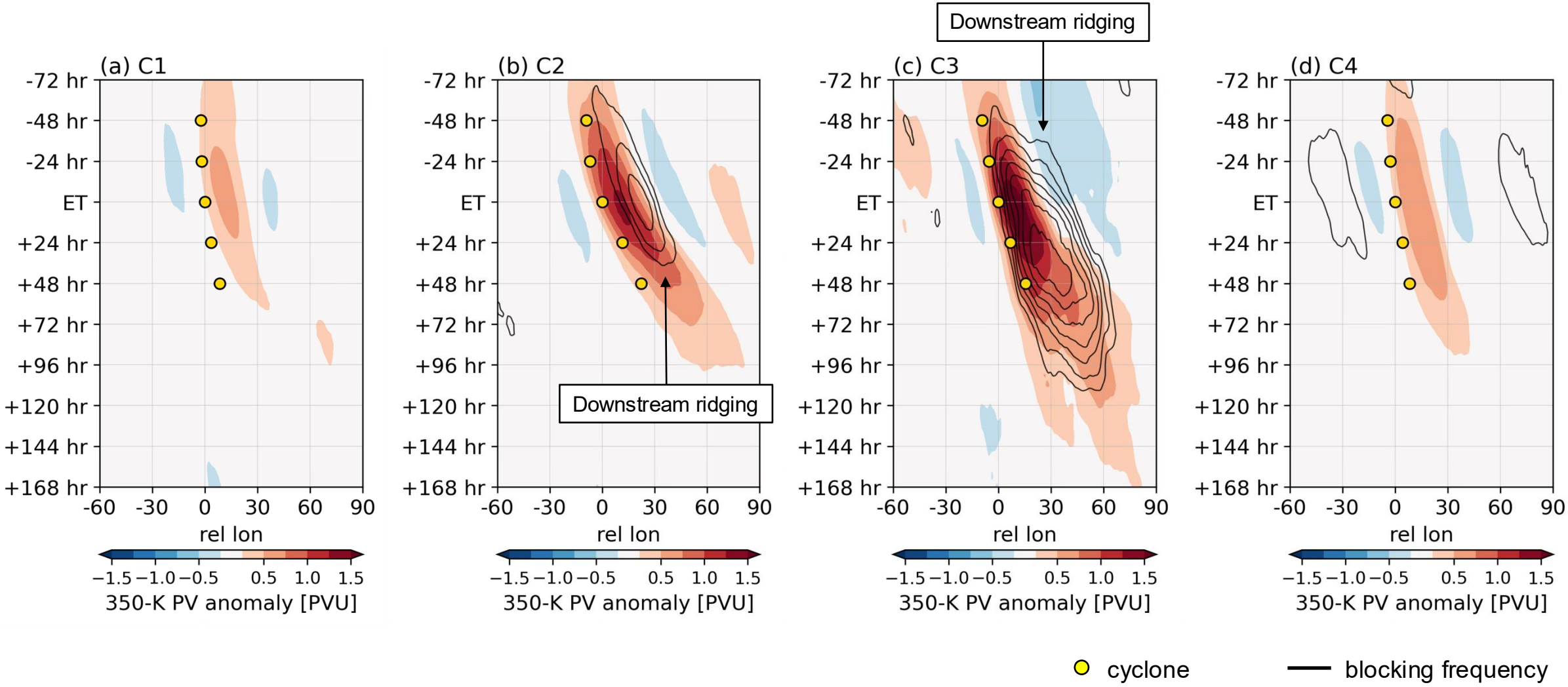


Synoptic patterns for ET clusters

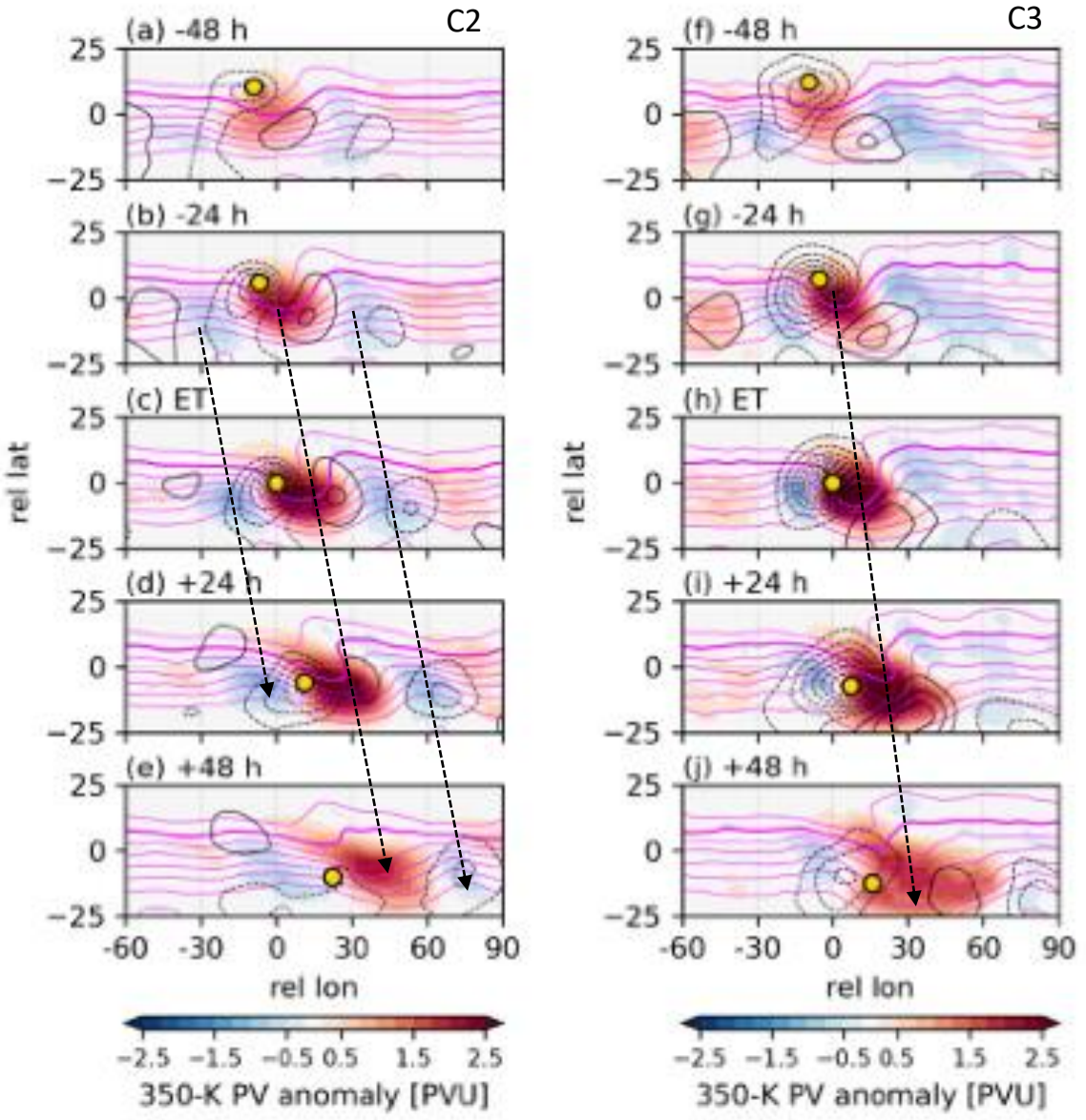
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Downstream development of ET clusters



Downstream development of ET clusters



- C2 features an eastward propagating Rossby wave packets and surface highs and lows
- C3 features a persistent strong downstream ridge

MSLP anomaly 350-K PV

PV budget

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Following Barnes et al. (2025), a PV budget is expressed as follows:

Propagation of Rossby wave

$$\frac{\partial q'}{\partial t} + \overline{\boldsymbol{V_\phi}} \cdot \nabla_\theta q' + \boldsymbol{V_\phi}' \frac{\partial \overline{q}}{\partial y} = -\boldsymbol{V_\chi} \cdot \nabla_\theta q - \boldsymbol{V_\phi}' \cdot \nabla_\theta q'$$

$$= I_\chi + I'_\phi$$

Interaction between diabatic heating and waveguide

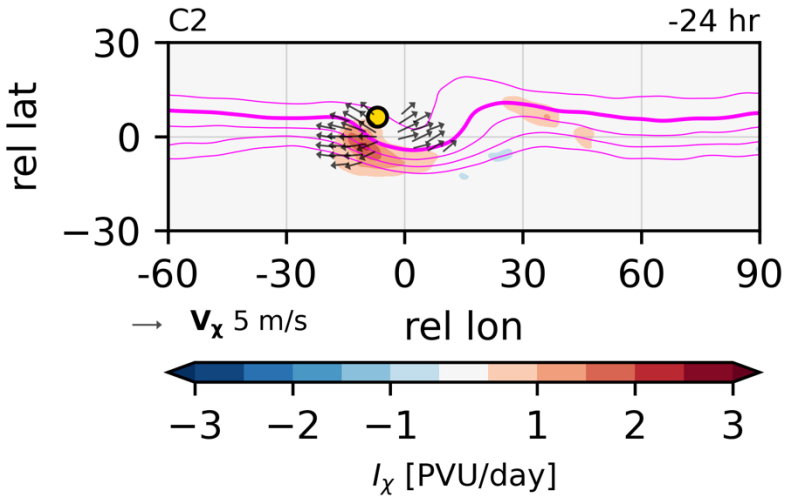
Nonlinearities

Where q' is PV anomaly from a time- and zonal-mean basic state, V_χ and V_ϕ represent the divergent and non-divergent part of horizontal wind.

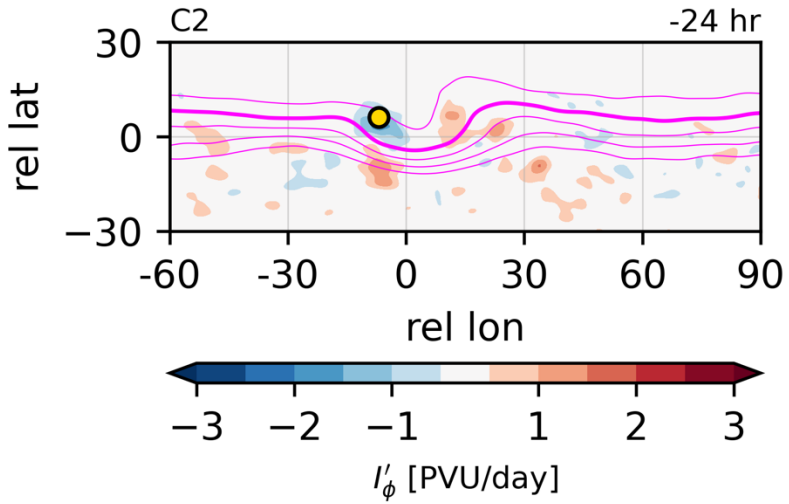
PV budget

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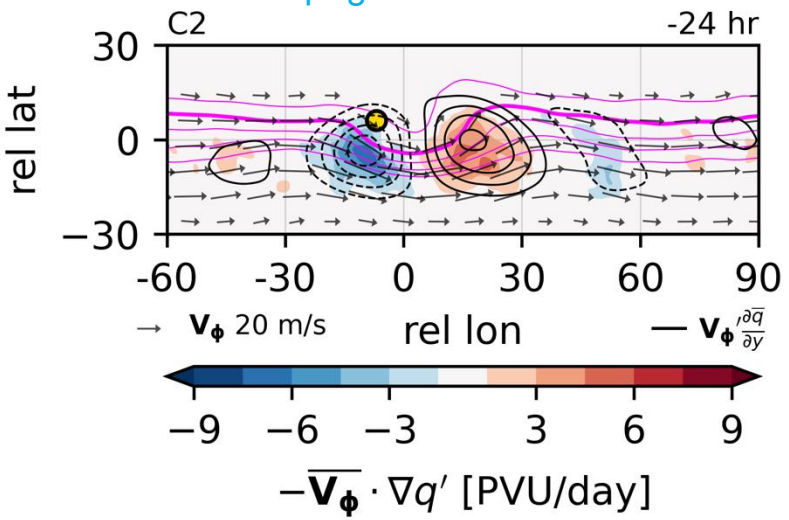
Diabatic effect

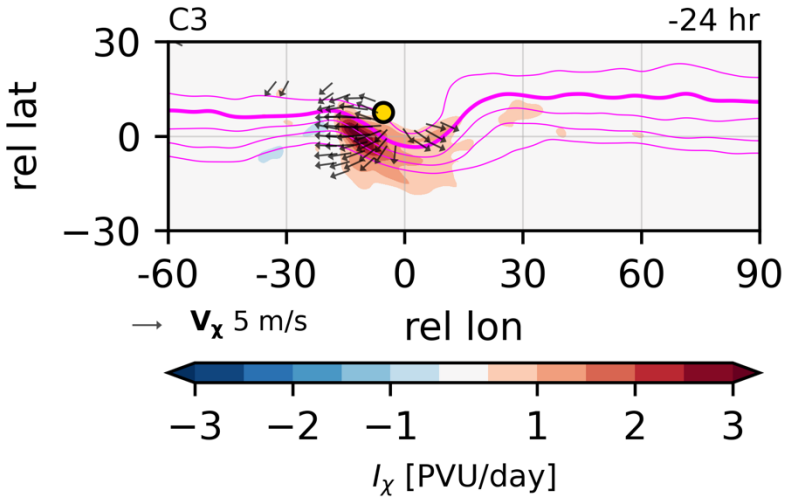


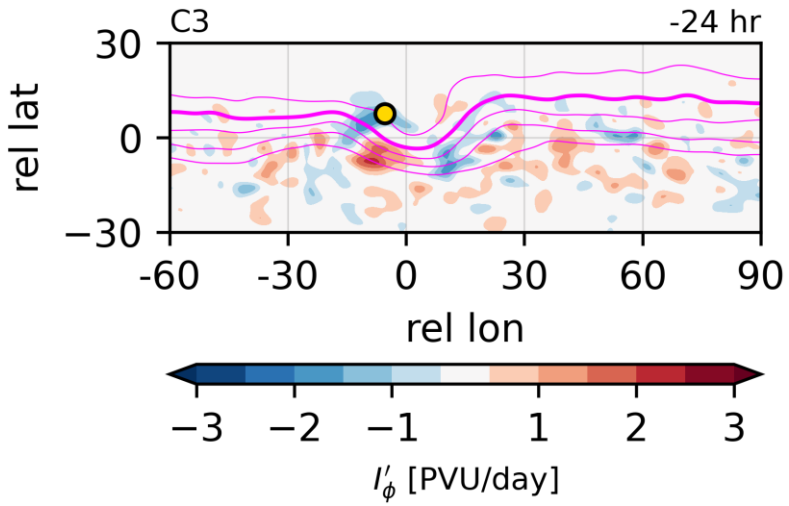
Nonlinearities

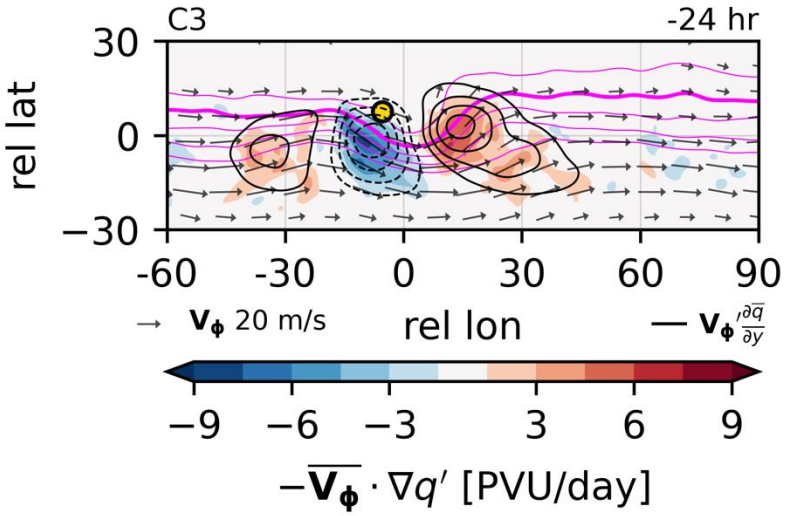


Propagation







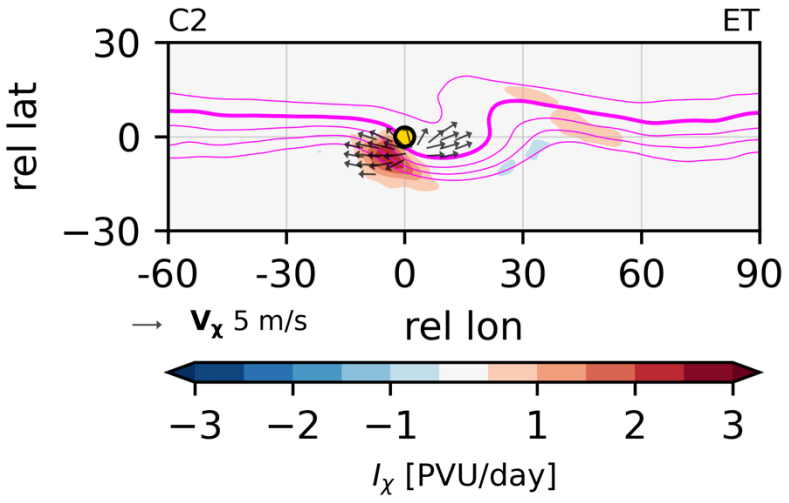


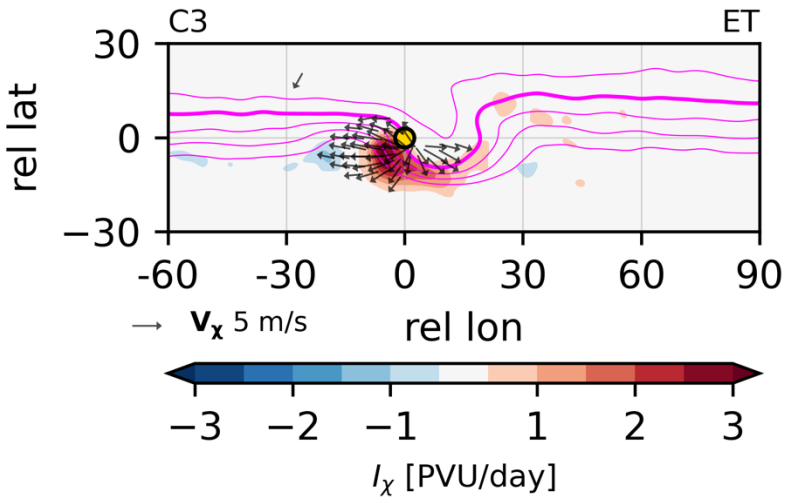
350-K PV

PV budget

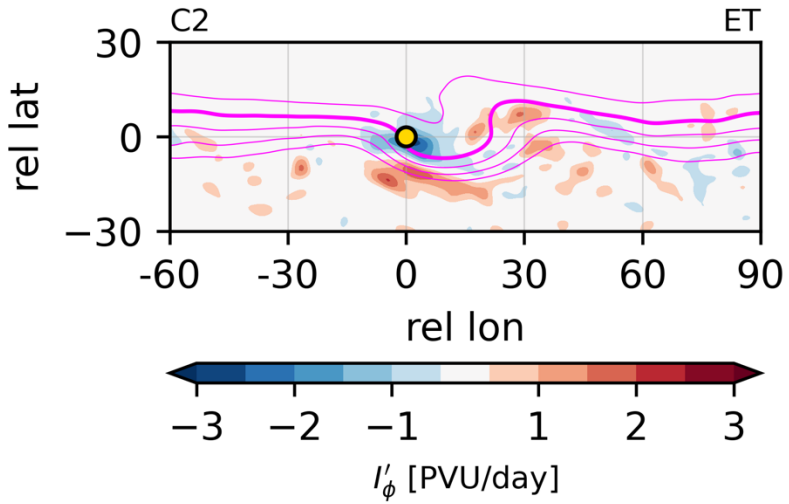
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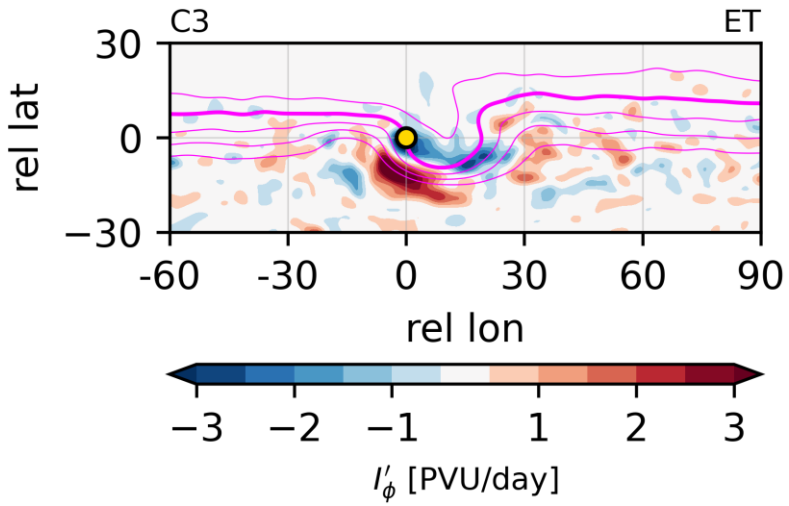
Diabatic effect



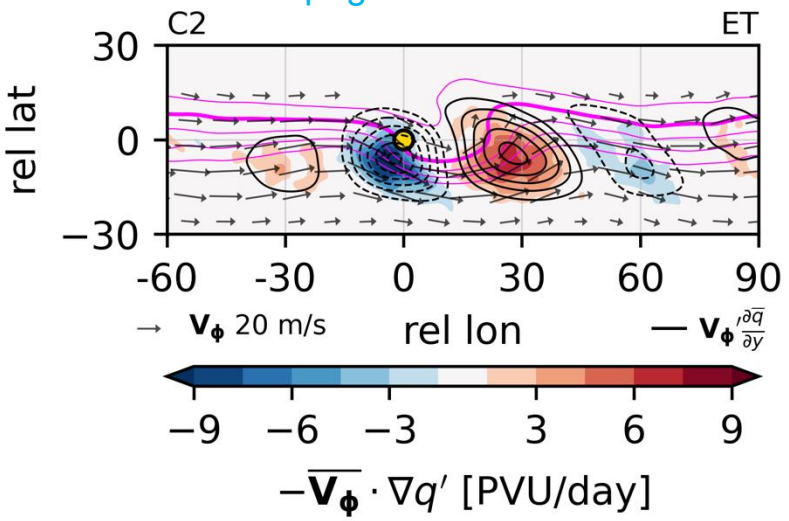


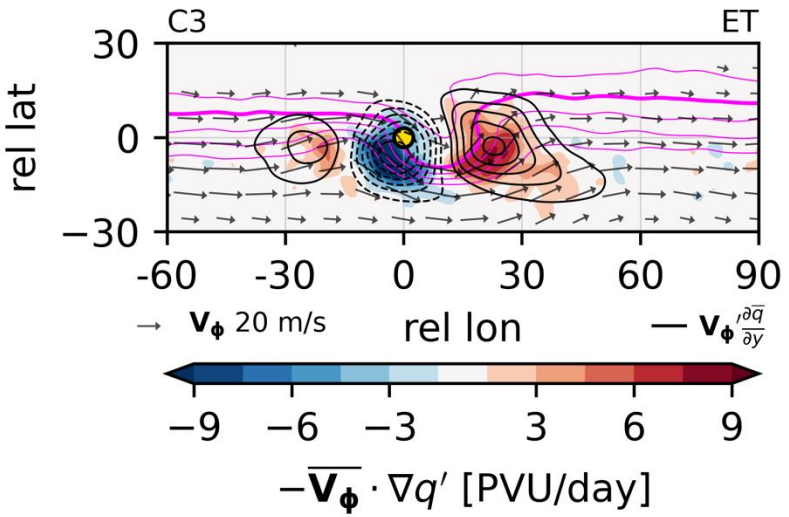
Nonlinearities





Propagation



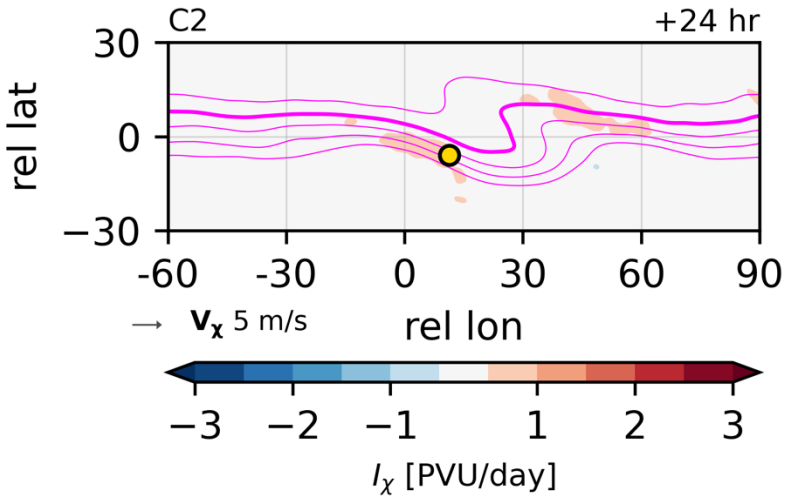


— 350-K PV

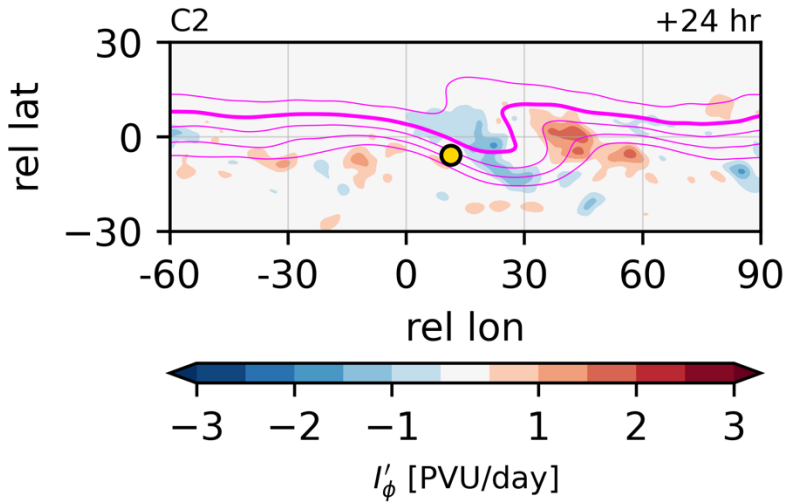
PV budget

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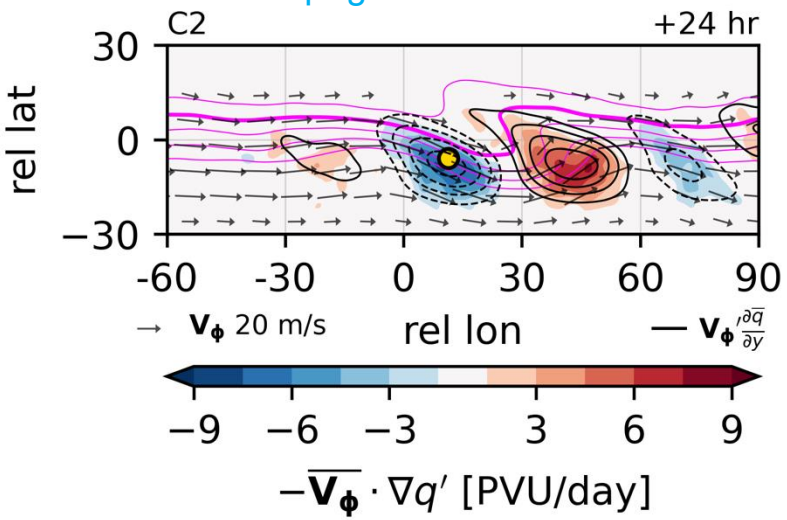
Diabatic effect

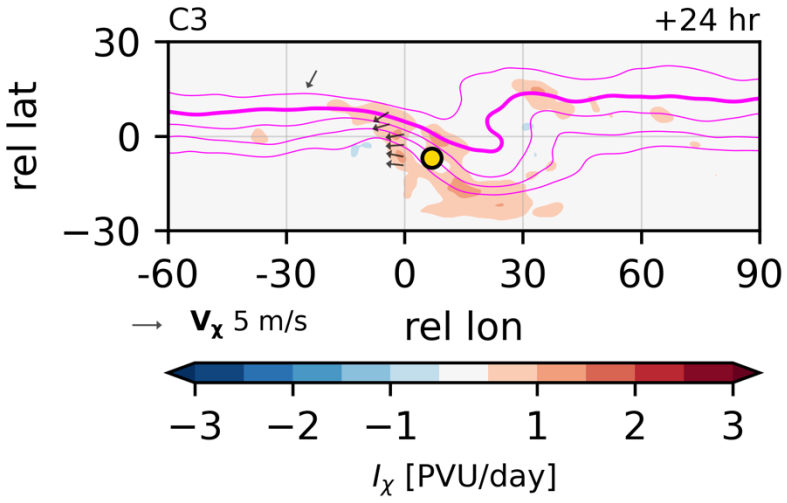


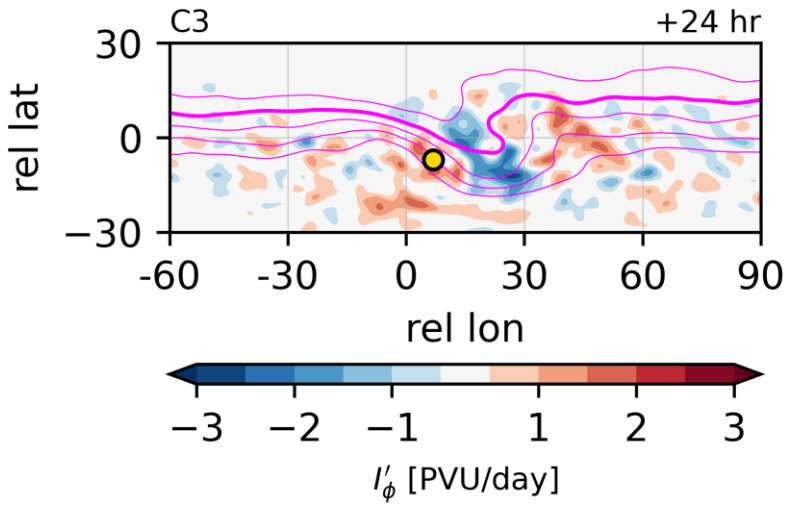
Nonlinearities

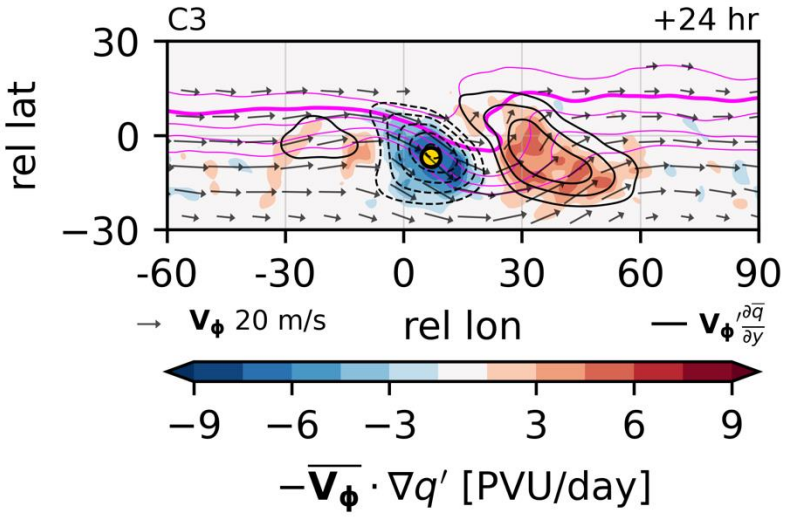


Propagation



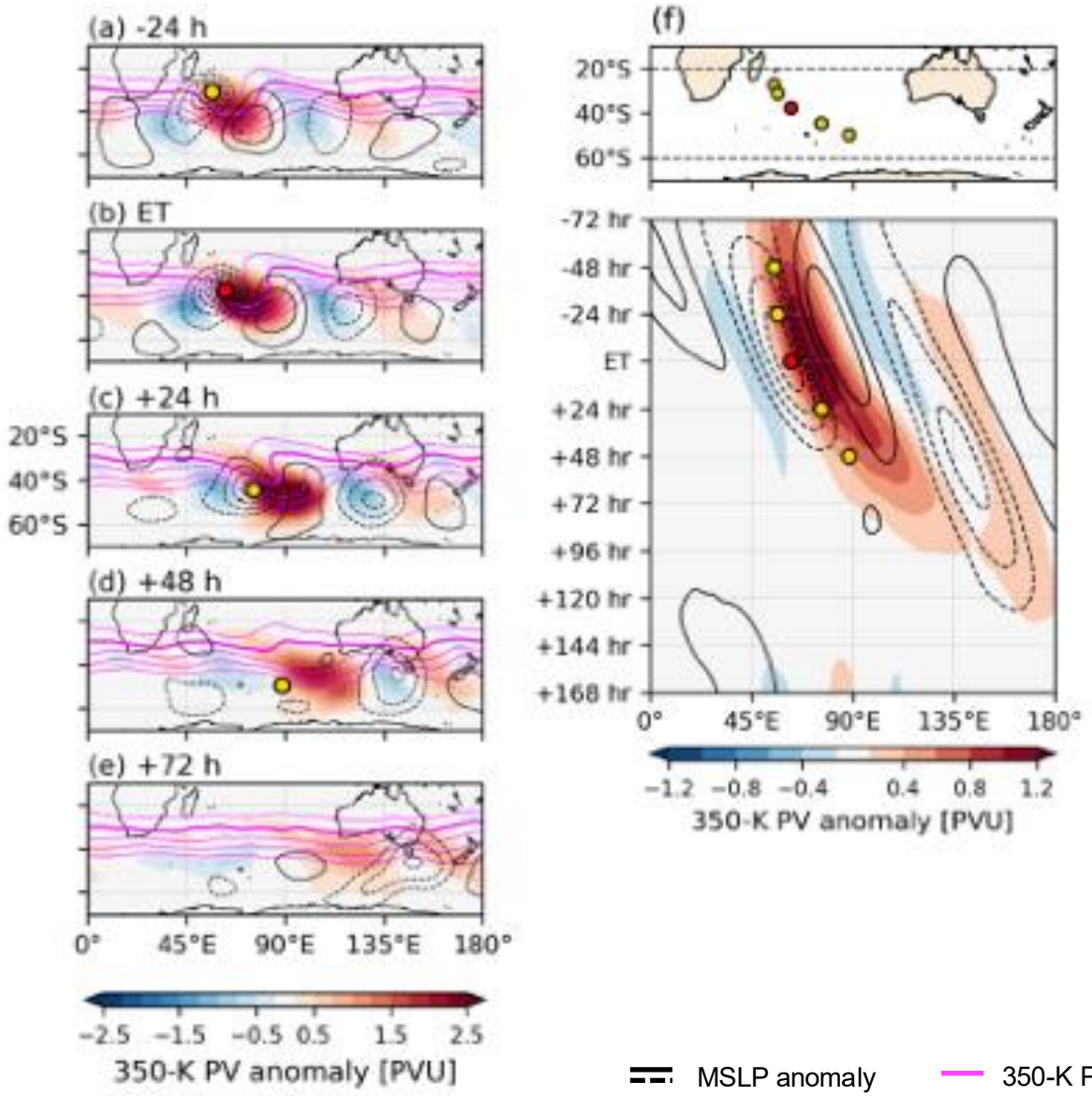






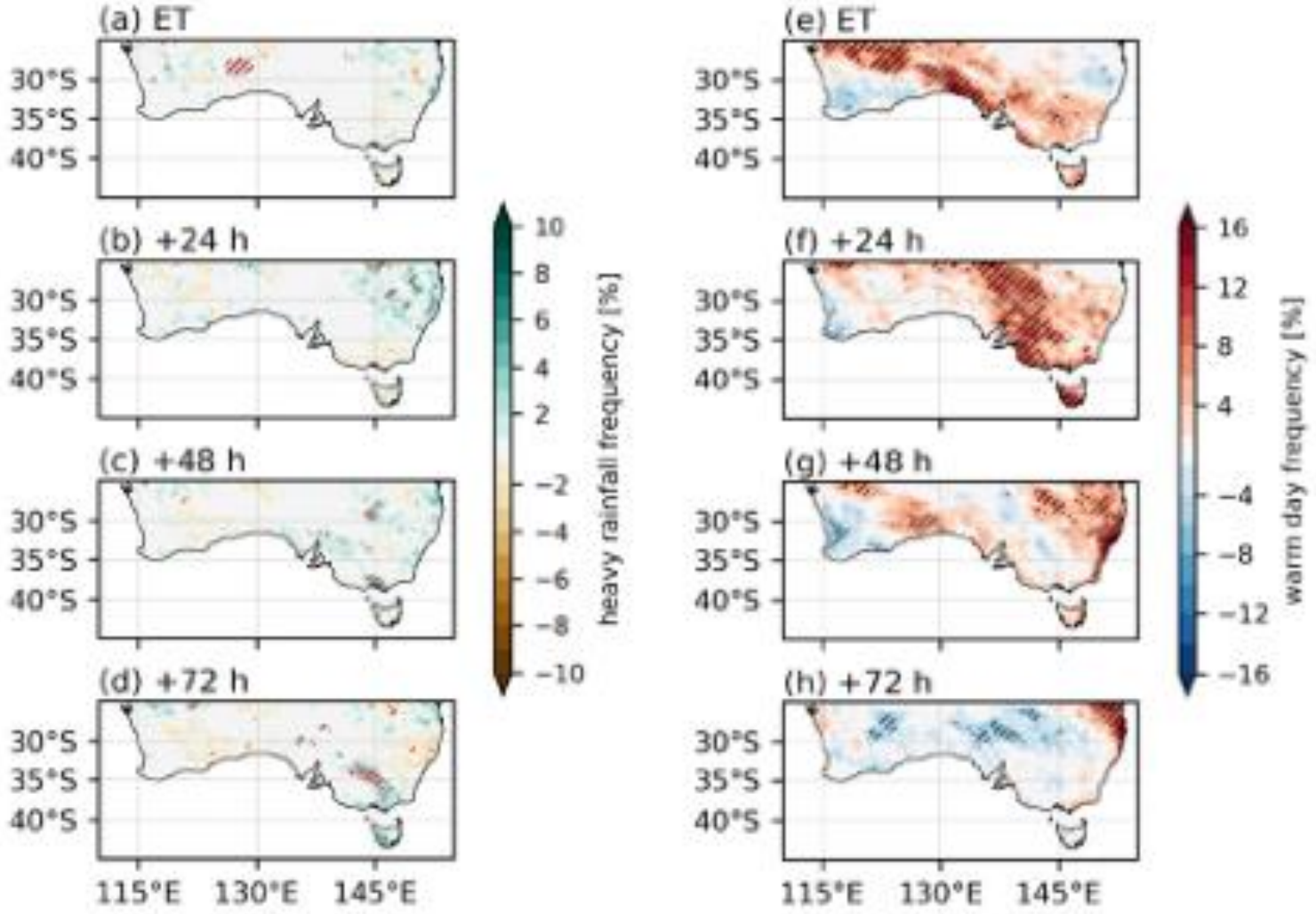
350-K PV

Downstream influences of ET over the southwest Indian Ocean



- A subset of C2 and C3 ET events over the southwest Indian Ocean
- Rossby wave packets develop downstream of ET, propagating toward Australia

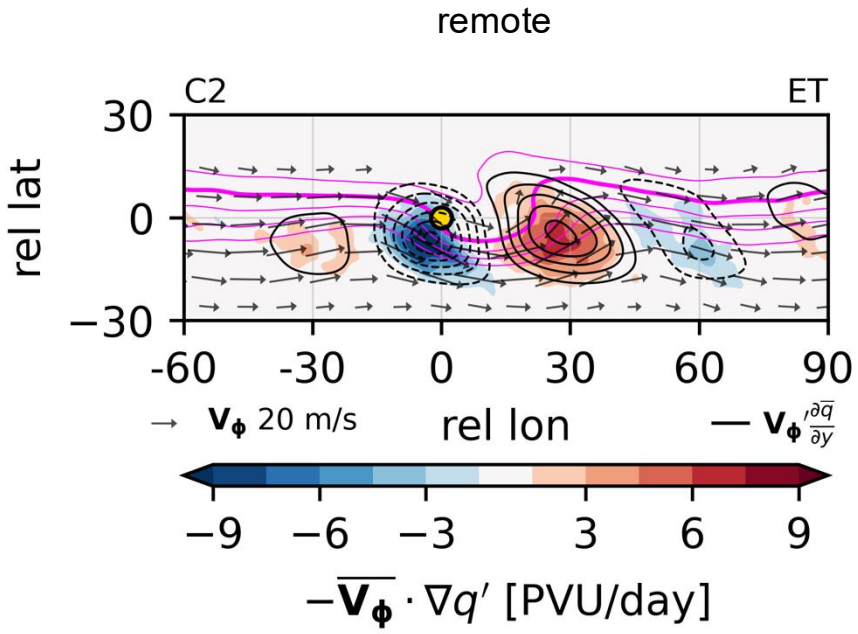
Downstream influences of ET over the southwest Indian Ocean

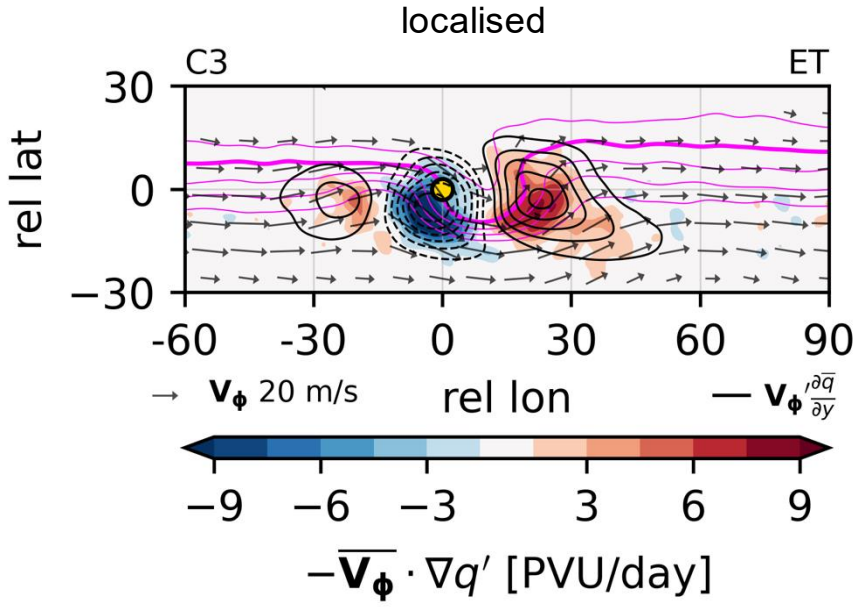


- Significantly increased heavy rainfall day frequency over the southeastern Australia 3 day after ET
- Significantly increased warm day frequency across the south to southeast Australia

Summary and future work

- Different pathways of downstream development (remote v.s. localised)





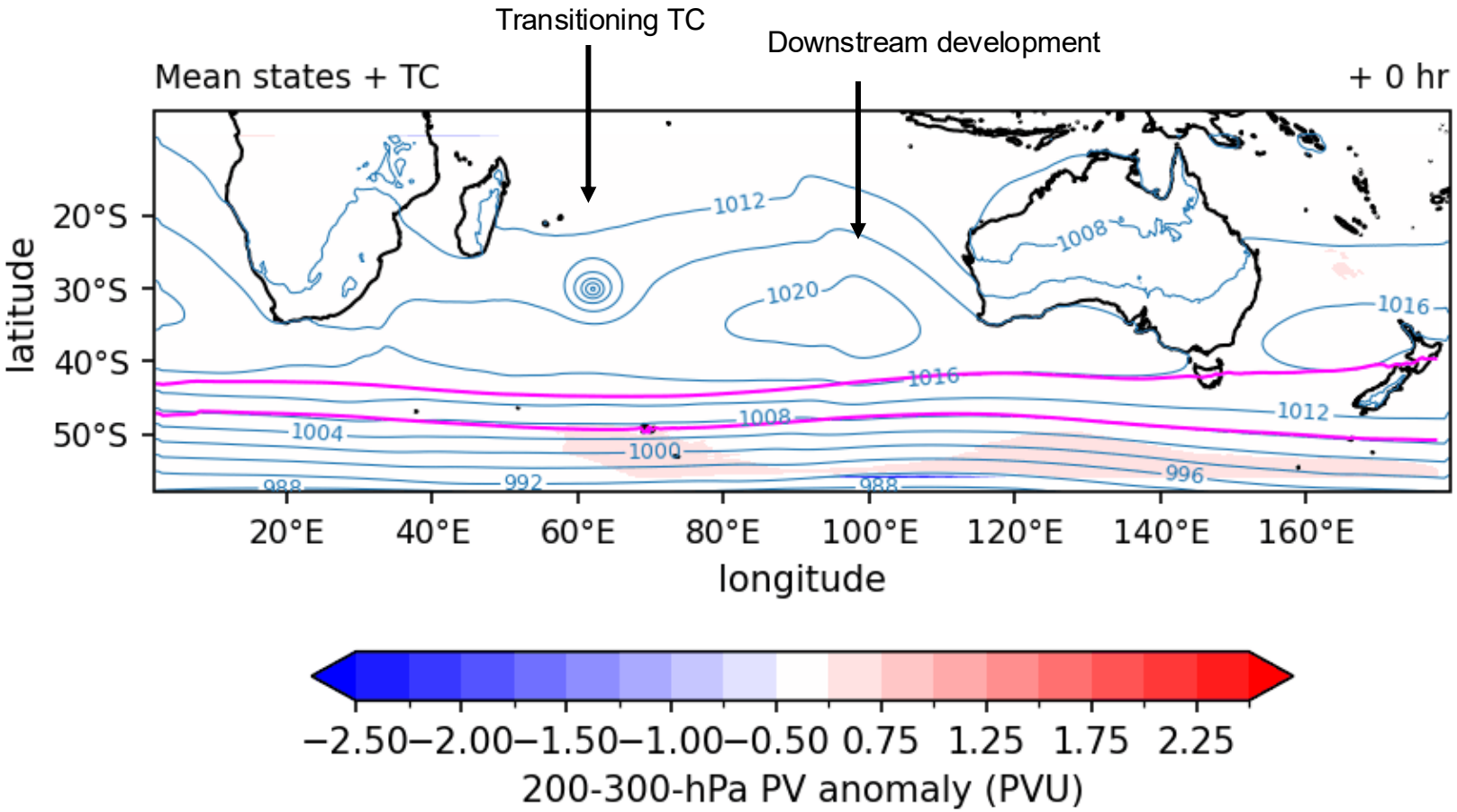
Jin et al. 2026 under revision in WCD

Jin et al. 2026 in prepare

Summary and future work

- Semi-idealised simulation on ET downstream development

  Perturbations associated with TC and upstream trough are added onto the climatological mean state, served as initial and boundary conditions



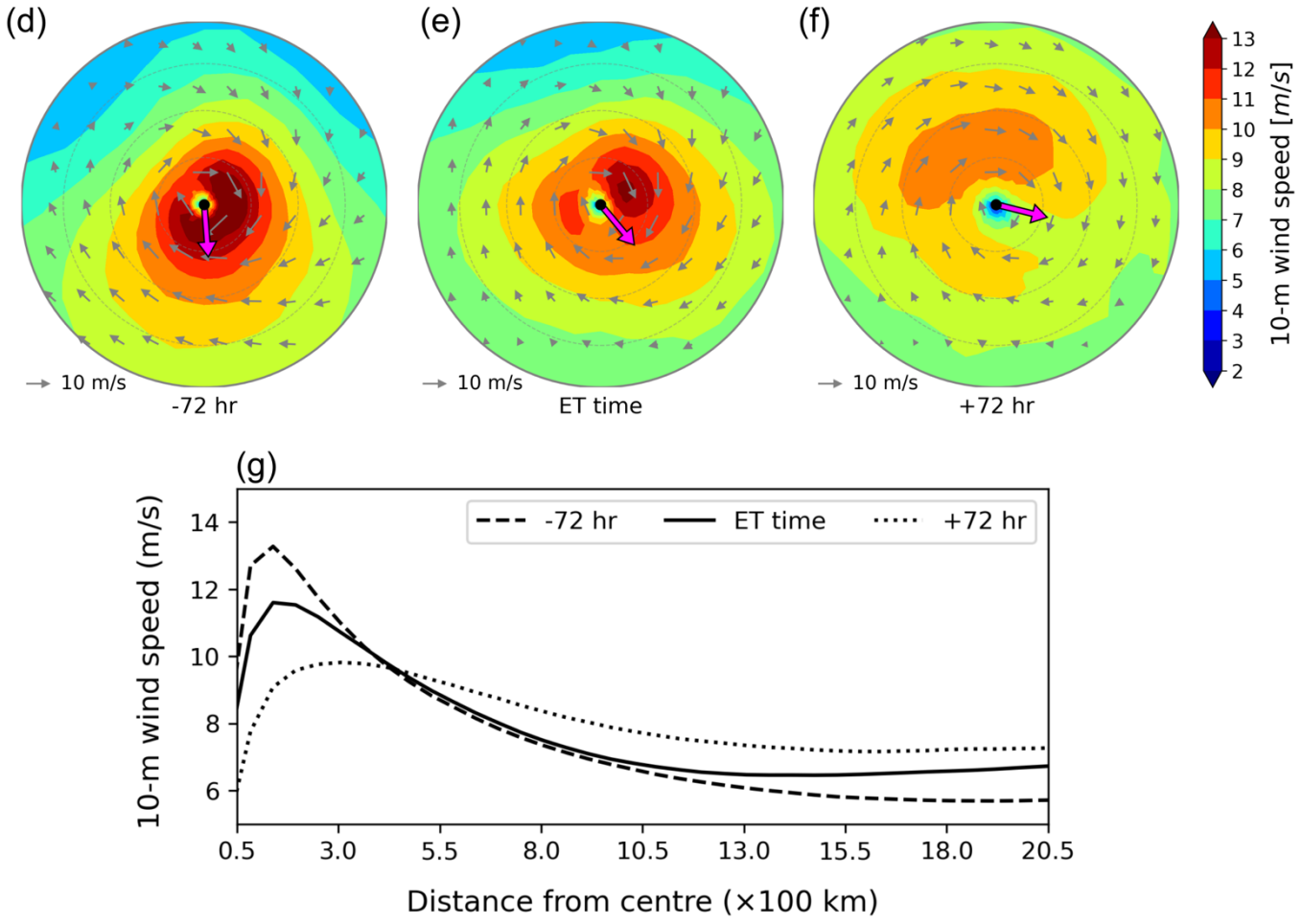
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Thank you

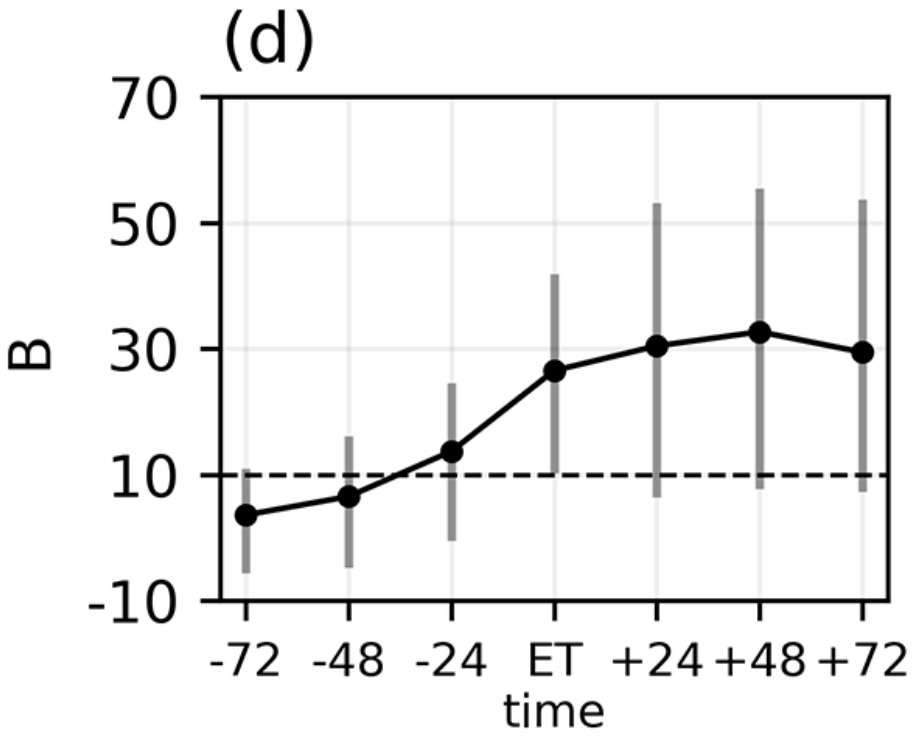


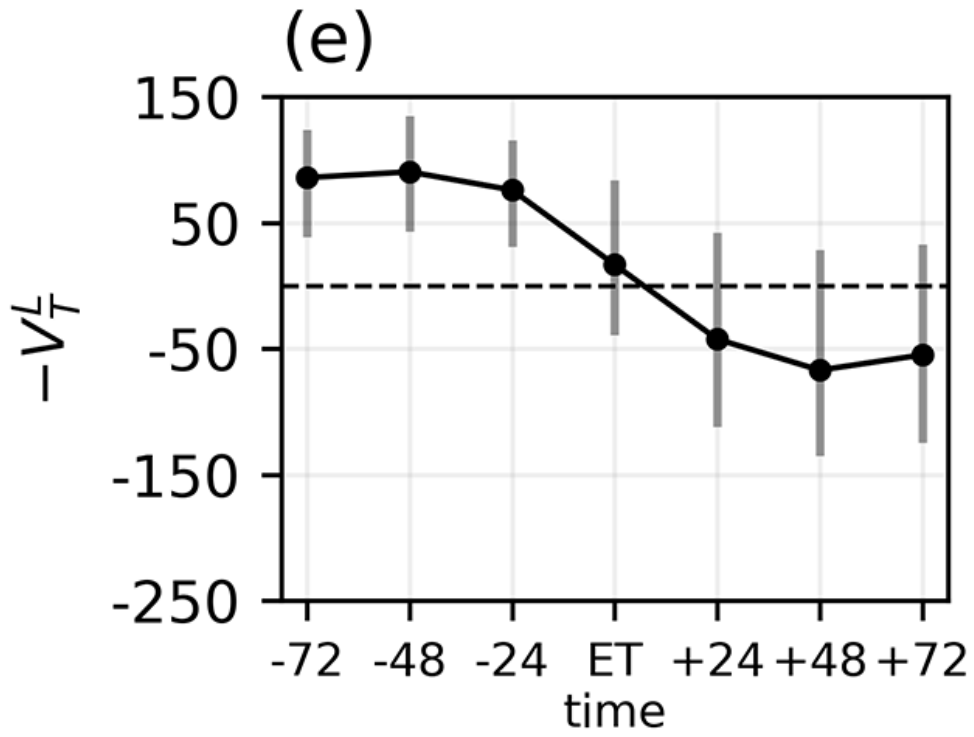
Appendix 1: wind field expansion



Wind field expands after ET

Appendix 2: CPS parameters





Eddy kinetic energy (Ke) budget

Following Orlanski and Sheldon (1995), the Ke budget is expressed:

ageostrophic geopotential eddy flux

Ke flux by the total wind

$$\frac{1}{g}\int_{1000}^{100}\frac{\partial K_e}{\partial t}\,dp=\frac{1}{g}\int_{1000}^{100}\left[-\nabla_p\cdot(\boldsymbol{v}\phi)_a-\omega\alpha-\nabla_p\cdot(\boldsymbol{V}K_e)-\boldsymbol{v}\cdot(\boldsymbol{v}\cdot\nabla_p\boldsymbol{V_m})+\text{residue}\right]dp$$

Baroclinic conversion

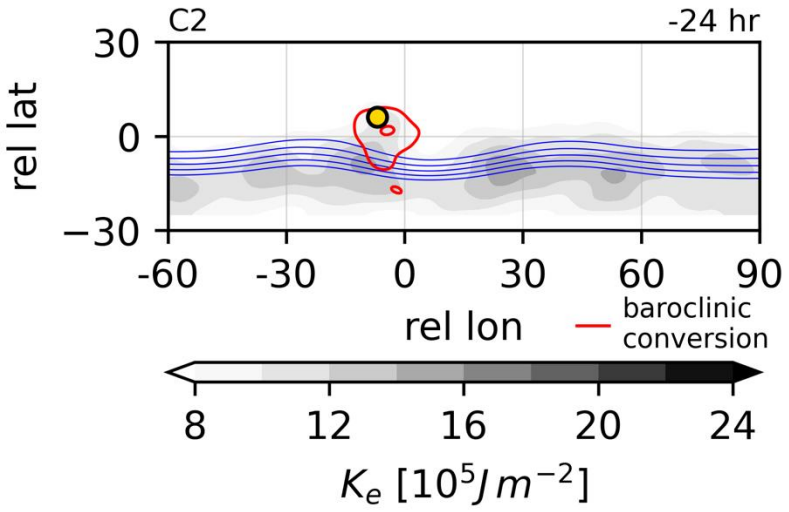
Barotropic conversion

Where the fields are partitioned as a sum of a time mean and a perturbation ($A = A_m + a$), $K_e = \frac{1}{2}|\boldsymbol{v}|^2$, ϕ is geopotential height perturbation, ω is vertical velocity perturbation, and α is the specific volume perturbation. A subscript a represent ageostrophic.

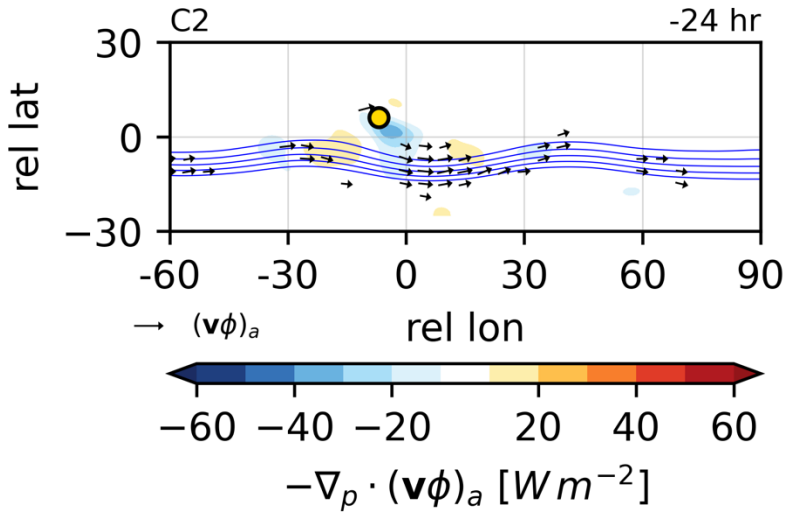
Ke budget

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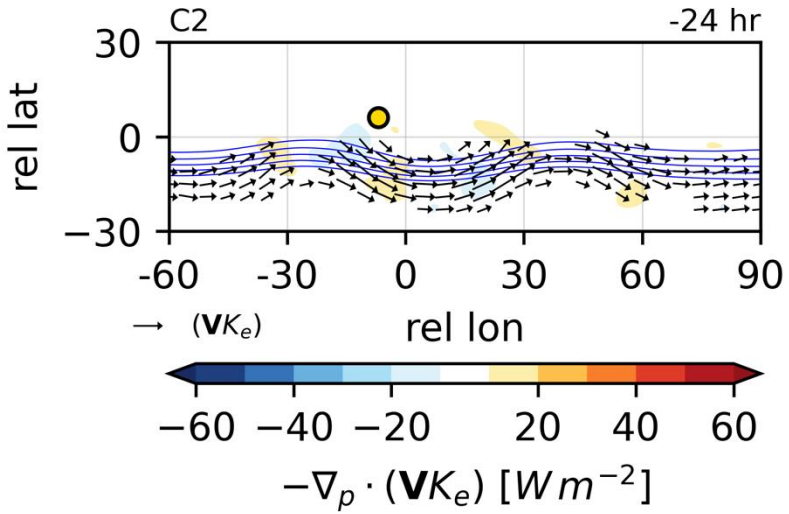
Baroclinic conversion

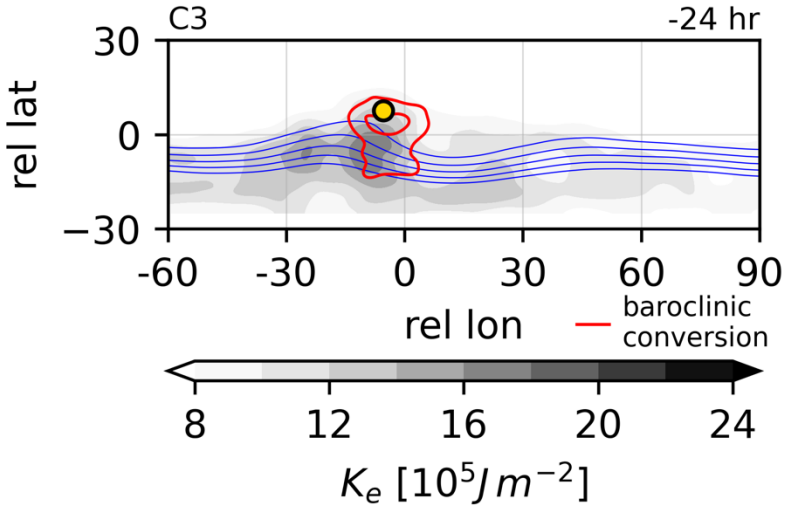


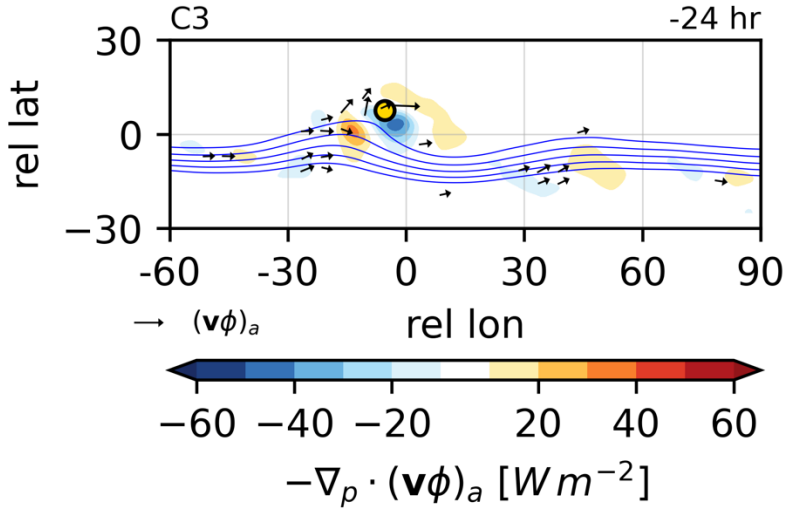
Ageostrophic geopotential eddy flux

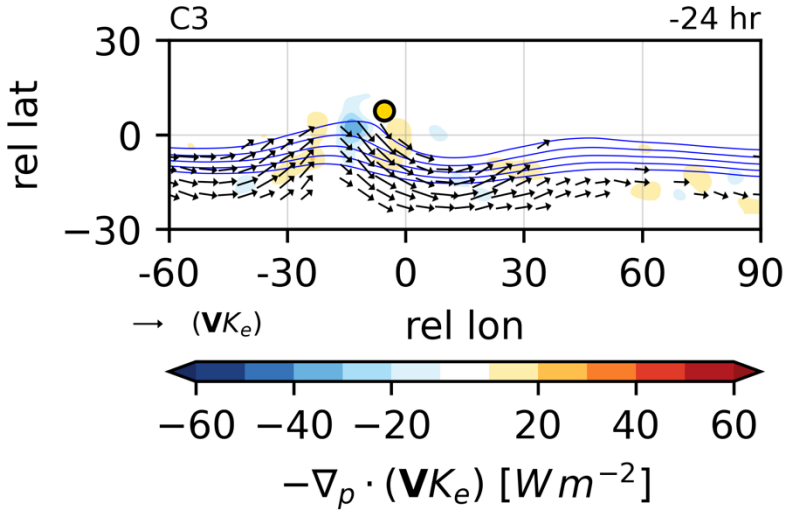


Ke flux by the total wind







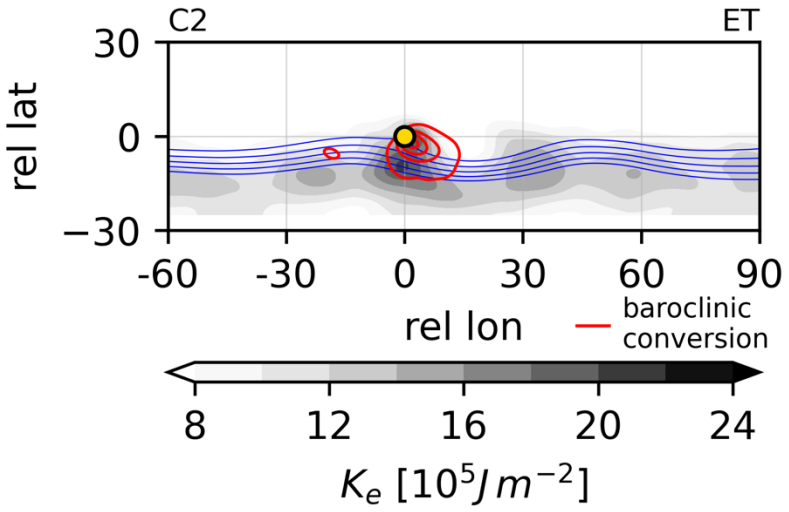


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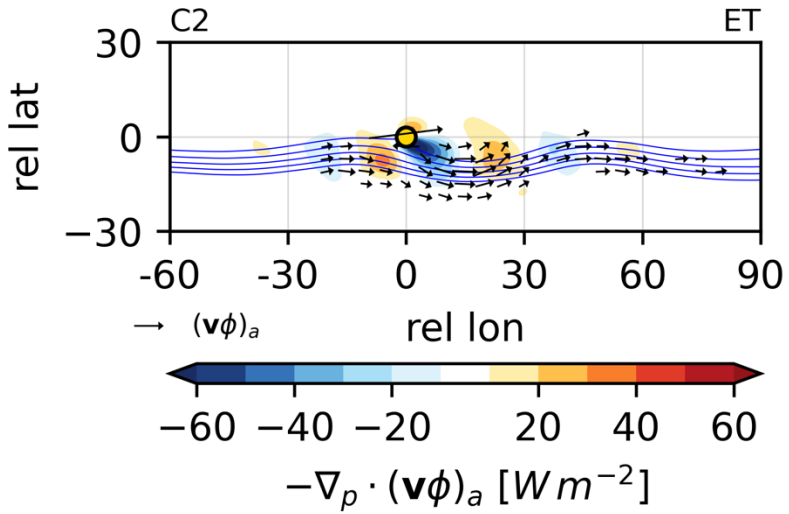
Ke budget

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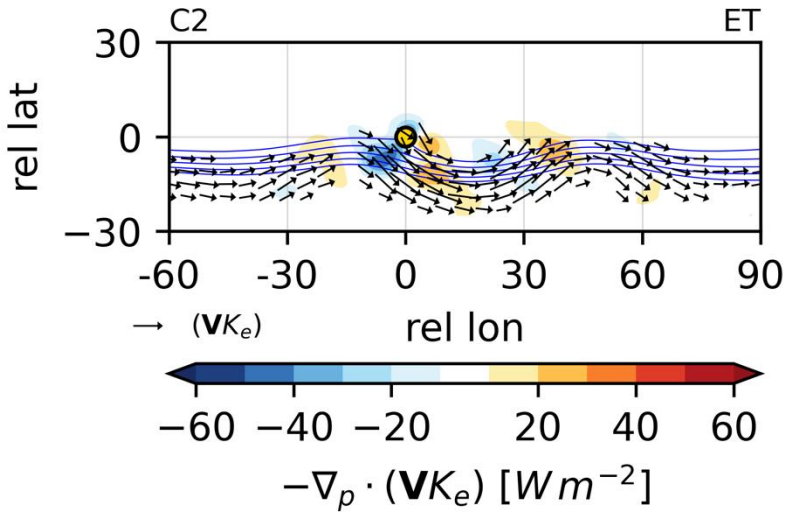
Baroclinic conversion

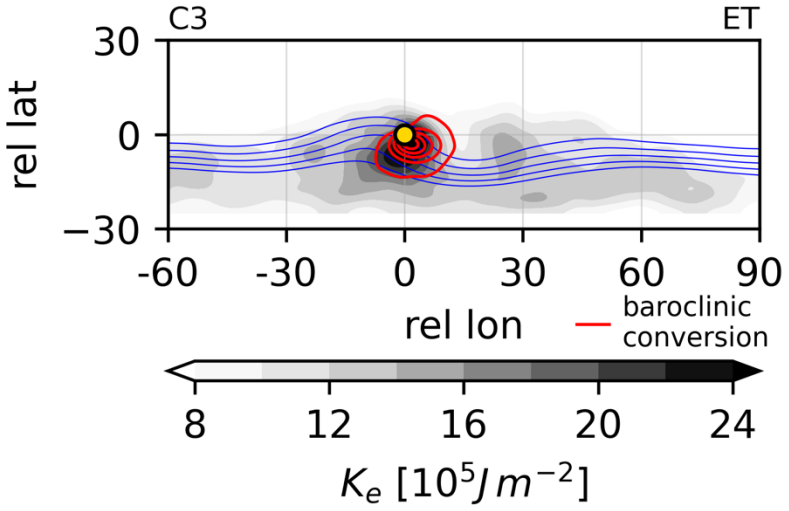


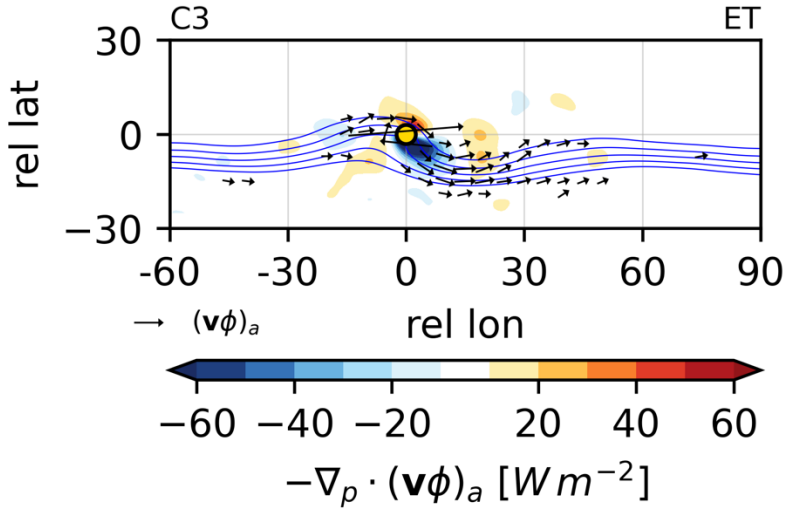
Ageostrophic geopotential eddy flux

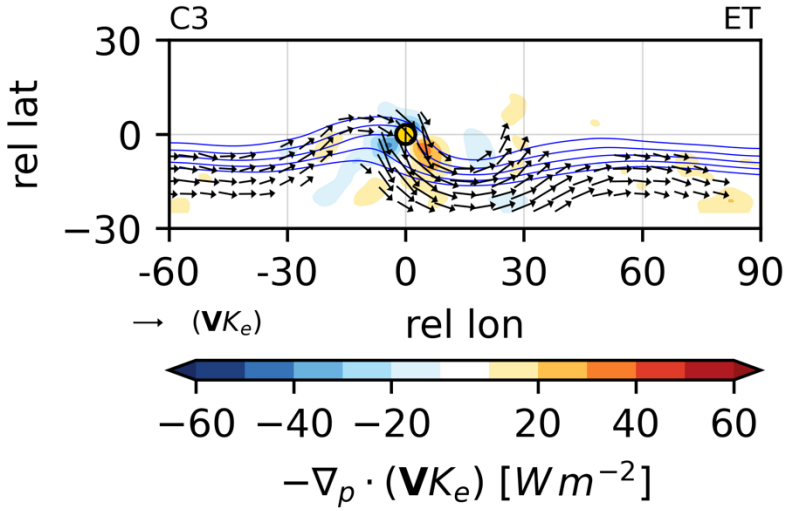


Ke flux by the total wind







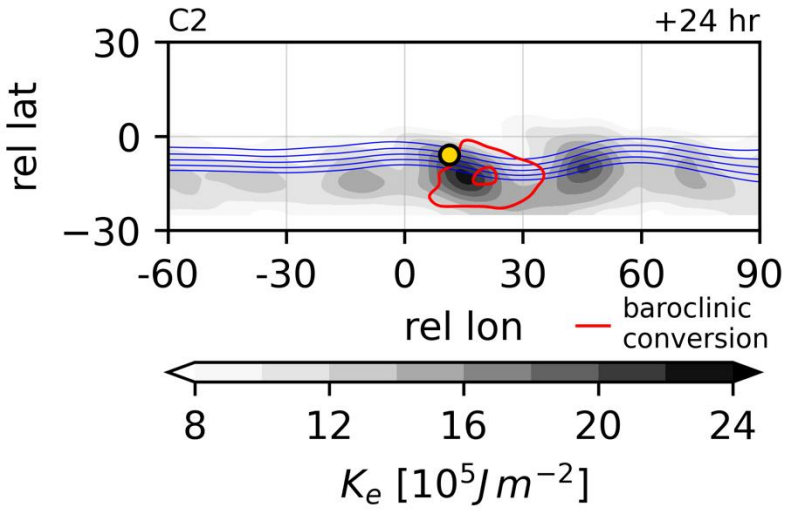


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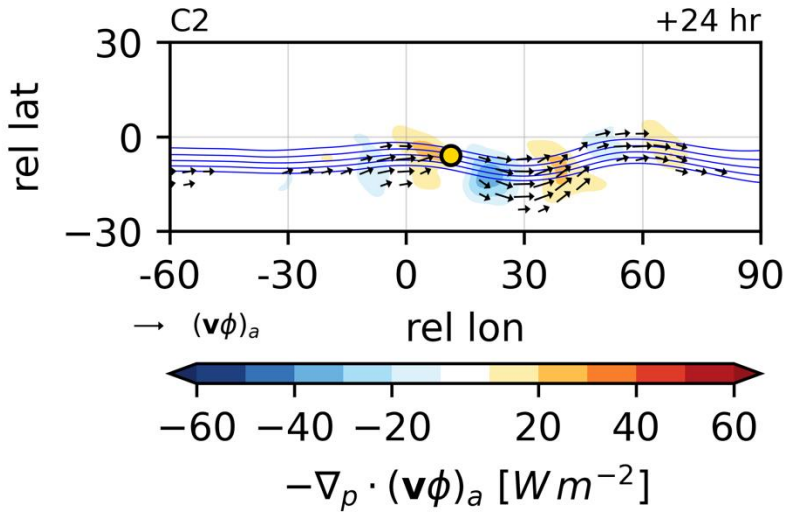
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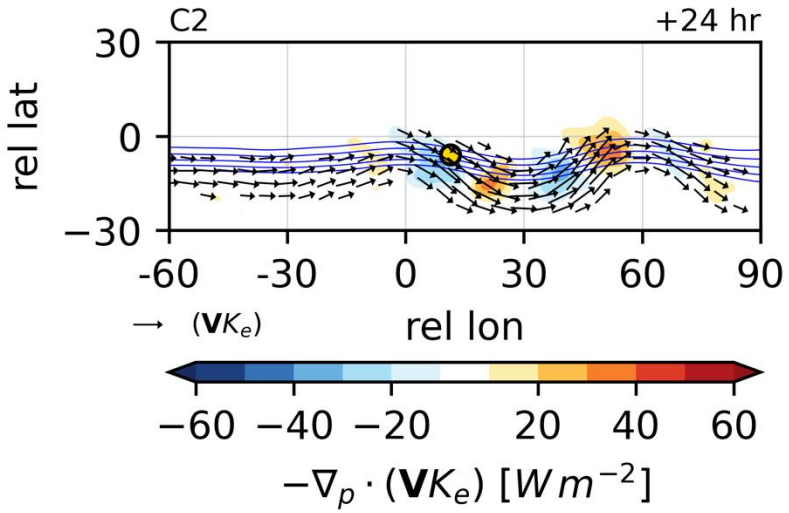
Baroclinic conversion

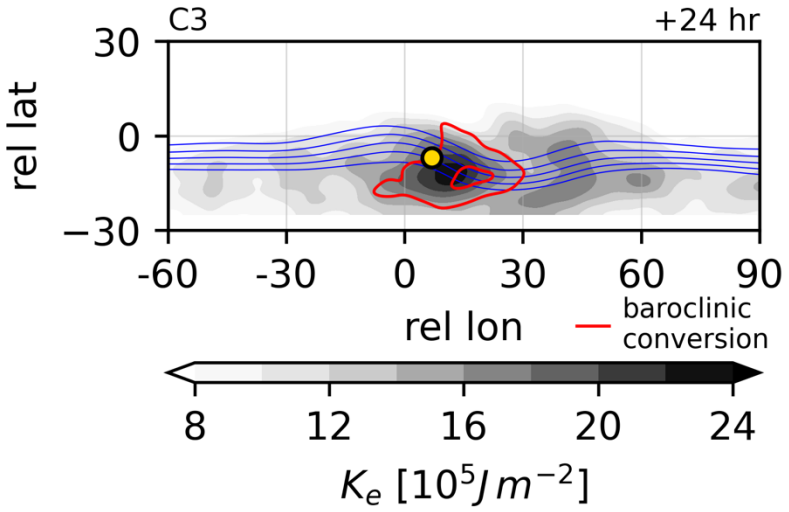


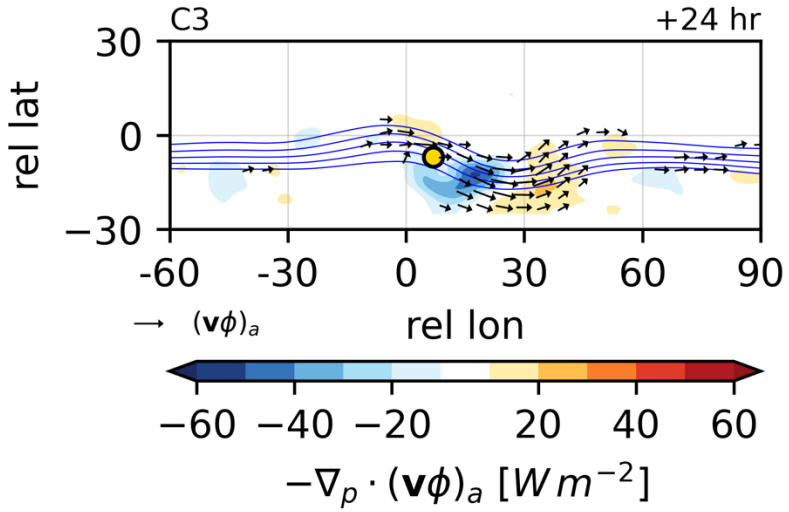
Ageostrophic geopotential eddy flux

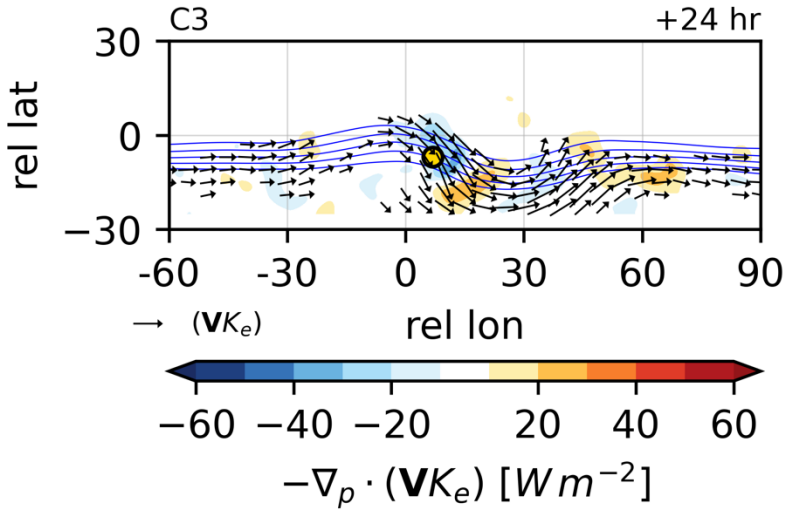


Ke flux by the total wind









Z500